HINDUSTHAN EDUCATIONAL AND CHARITABLE TRUST

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INSTITUTE OF TECHNOLOGY

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DEPARTMENT OF MANAGEMENT SCIENCES

20PBA09 OPERATIONS MANAGENENT

Department	MASTER OF BUSINESS ADMINISTRATION	R 2022	Sem.	II

Course Code	Course Name	Hours / Week			Credit	Total Hours	Maximum Marks
		L	T	P	C		
22PBA09	OPERATIONS MANAGEMENT	3	0	0	3	45	100

Course Objective

1. To understand the importance of operations management to survive in the business
2. To learn the different systems involved in manufacturing resource planning
3. To practice the various method involved in production planning
4. To evaluate the various techniques in materials management practices
5. To realize the importance of quality for the business and the society

UNIT I – INTRODUCTION TO OPERATIONS MANAGEMENT **9 Hours**

Operations Management (OM)–Nature, Importance, historical development, transformation processes – Functions of OM - Differences between services and goods - system view of operations management -Types of production system – Operations Strategy &Strategic management -Supply Chain Management

UNIT II – MANUFACTURING RESOURCE PLANNING **9 Hours**

Demand Forecasting– Need, Types, Objectives and Steps .Plant location – Need , importance & factors influencing . Facility layout – Importance, types. Factors to be considered – Capacity planning – Simple problems – Make or Buy decision – Problems. Product Design – Stages.

UNIT III - PRODUCTION PLANNING **9 Hours**

Aggregate Planning Strategies- Types – Functions of production planning and control – Process planning – steps and functions - Work study – Method study & Work measurement – Scheduling & Shop floor control – John‘s rule – PERT & CPM Techniques – Simple problems – Value analysis

UNIT IV - MATERIALS MANAGEMENT **9 Hours**

Materials requirement planning (MRP – I) , Objectives & functions, Simple problems – Purchasing procedure, functions & methods –Vendor rating – Stores management objectives, functions – Classification & coding – Types of stores – Inventory control Objectives & functions – Safety stock, Re-order level – Inventory control techniques – EOQ & EBQ models , simple problems. JIT, KANBAN

UNIT V - QUALITY MANAGEMENT **9 Hours**

Quality – Definitions & dimensions of quality – Evolution of quality – Total quality management (TQM) – TQM frame work & implementation – Statistical Quality Control – Acceptance sampling , Types – Operation Characteristics Curve – Control charts X, P & C Chart – Problems – Process capability – Six sigma – Quality management tools Old & New. - Quality Circle – An overview of ISO 9000 & IS 14000 system – Role of information management in Operations

Problem 20 % **TOTAL HOURS: 45 PERIODS**

State of the Art	**Basics of Operations Management**	
Course Outcome	CO1	Able to craft an operations strategy in a dynamic environment
	CO2	Capable of utilizing the manufacturing resources effectively
	CO3	Capacity to schedule the production to deliver the products beyond customer Expectations
	CO4	Acquired the materials management practices towards efficiency
	CO5	Proceed in the business with focus on quality

TEXT BOOK

1. Aswathappa K and Shridhara Bhat K, ―Production and Operations Management‖, Himalaya Publishing House, 3rd edition, 2015.

REFERENCE BOOKS

1. Richard B. Chase, RaviShankar, F. Robert Jacobs, Nicholas J. Aquilano, ―Operations and Supply Chain Management‖, Tata Mc Graw Hill, 12th edition, 2010.
2. Norman Gaither and Gregory Frazier, ―Operations Management‖, South Western Cengage, 9th edition, 2006.
3. Pannerselvam R, ―Production and Operations Management‖, Prentice Hall India, 3rd edn , 2012
4. Russel and Taylor, ―Operations Management‖, Wiley, 8th Edition, 2015.
5. Mahadevan B, ―Operations Management - Theory and practice‖, Pearson Education, 3rd Edition, 2015

WEB REFERENCES:

1.https://nptel.ac.in/courses/110/107/110107141/
2.https://nptel.ac.in/courses/110/107/110107116/

CO PO MAPPING

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	1	3	1	1	1	1		1		1	1	1	1
CO2	2	2	2	3	2	1	1		1		2	1	2	1
CO3	2	2	2	2	1	1	1		1		1	1	2	1
CO4	2	2	2	2	1	1	1		2		1	1	2	1
CO5	2	1	1	2	2	2	1		2		1	1	2	1

HINDUSTHAN INSTITUTE OF TECHNOLOGY, COIMBATORE – 32.

DEPARTMENT OF MANAGEMENT SCIENCES

BA 7201 OPERATIONS MANAGEMENT

UNIT I INTRODUCTION TO OPERATIONS MANAGEMENT

Operations Management – Nature, Importance, historical development, transformation processes, differences between services and goods, a system perspective, functions, challenges, current priorities, recent trends; Operations Strategy – Strategic fit , framework; Supply Chain Management

Introduction:

Operation is that part of as organization, which is concerned with the transformation of a range of inputs into the required output (services) having the requisite quality level. Management is the process, which combines and transforms various resources used in the operations subsystem of the organization into value added services in a controlled manner as per the policies of the organization. The set of interrelated management activities, which are involved in manufacturing certain products, is called as production management. If the same concept is extended to services management, then the corresponding set of management activities is called as operations management.

PRODUCTION:

Production function is _the part of an organisation, which is concerned with the transformation of a range of inputs into the required outputs (products) having the requisite quality level‘.

Production is defined as the step-by-step conversion of one form of material into another form through chemical or mechanical process to create or enhance the utility of the product to the user‘.

Thus production is a value addition process. At each stage of processing, there will be value addition. It implies the creation of goods and services to satisfy the human wants

According to Card Heyel, Production is the process of transforming raw material and component parts in to finished product for sale.

PRODUCTON MANAGEMENT:

It is concerned with planning, organizing, directing and controlling of production system so as to produce goods of desired quality at minimum cost and the right time. It deals with decision making concerning production processes.

According to E.L.F.Brech, ―Production Management then becomes the process of effectively planning, and regulating the operations of that an enterprise which is responsible for all actual transformation of materials into finished products‖

OPERATION MANAGEMENT:

It involves the three functions of getting inputs, conversion and outputs. **Inputs** consist of people, material and equipment. The **conversion** activity consists of design fabrication, assembly and storage. The **output** comprises goods and services

Operation management means the conversion of inputs into outputs involves design and development of products and services, facilities location, capital equipment, facilities layout, work design and measurement, production forecasting, production planning and scheduling, material management, inventory management and quality control.

Objectives of Production and operation management:

The objective of the production management is _to produce goods and services of Right Quality and Quantity at the Right time and Right manufacturing cost‘.

1. Right Quality: The quality of product is established based upon the customers need. The right quality is not necessarily being the best quality. It is determined by the cost of the product and the technical characteristics as suited to the specific requirements.

2. Right Quantity: The manufacturing organisation should produce the products in right number. If they are produced in excess of demand the capital will block up in the form of inventory and if the quantity is produced in short of demand, leads to shortage of products.

3. Right Time: Timeliness of delivery is one of the important parameter to judge the effectiveness of production department. So, the production department has to make the optimal utilization of input resources to achieve its objective.

4. Right Manufacturing Cost: Manufacturing costs are established before the product is actually manufactured. Hence, all attempts should be made to produce the products at pre-established cost, so as to reduce the variation between actual and the standard (pre-established) cost.

⇒ It aims at manufacturing high quality products at minimum cost.

⇒ It eliminates duplication of work and overlapping efforts, leading to increased efficiency.

⇒ It increases productivity.

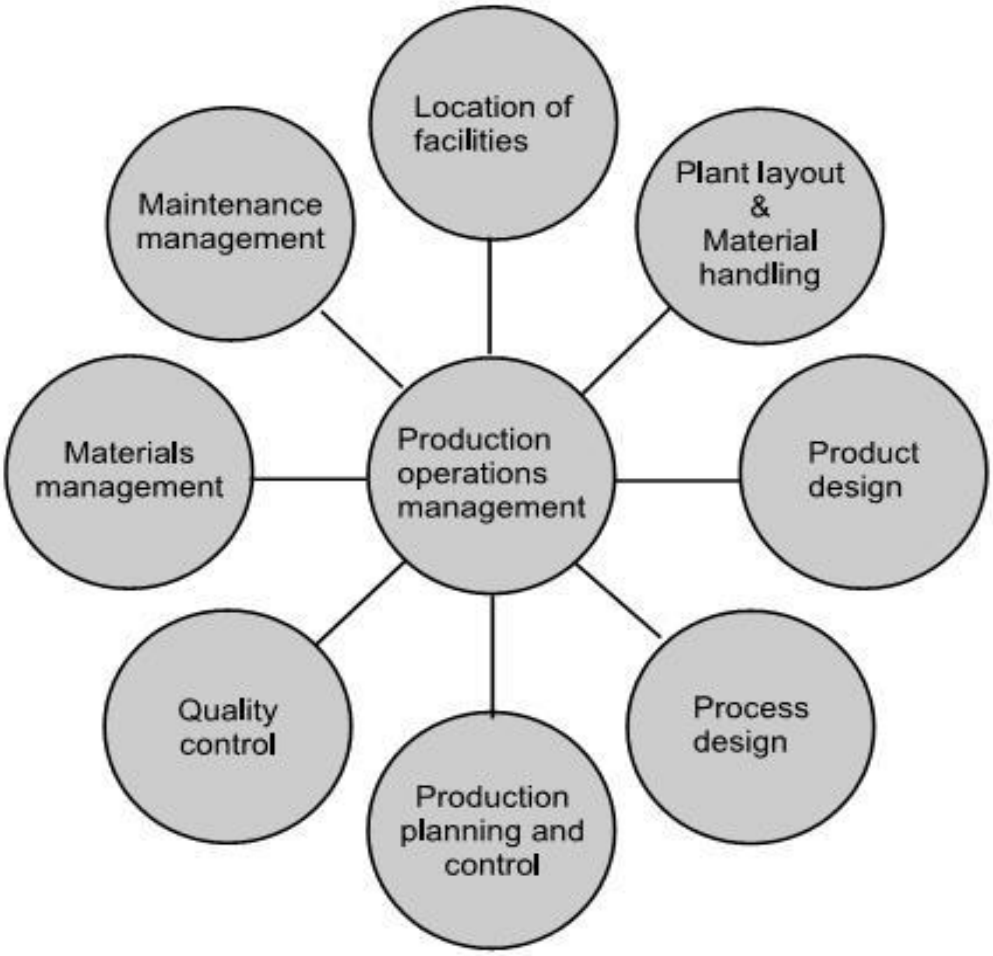
⇒ It leads to higher efficiency which, in turn, increases profit.

⇒ It minimizes the cost of production which generates more revenue to the company.

⇒ It facilitates for product planning and development which results in the development to new product.

SCOPE OF OPERATIONS MANAGEMENT

Operations Management concern with the conversion of inputs into outputs, using physical resources, so as to provide the desired utilities to the customer while meeting the other organizational objectives of effectiveness, efficiency and adoptability. It distinguishes itself from other functions such as personnel, marketing, finance, etc. by its primary concern for _conversion by using physical resources‘. Following are the activities, which are listed under Production and Operations Management functions:



1. Product planning and development

Product planning deals with determination of product for manufacturing purpose.

What to produce?

What quality to produce?

How much to produce?

When to produce?

This product planning is done by the production management is concerned with marketing management. Product development deals with designing a new product. Every product design determinates a success or failure. Hence much care should be strength in product designing. Before designing the product standardization is considered.

2. Plant Location:

Plant location deals with determination of suitable area for locating the plant. In order to increase the efficiency of the production department the plan should be located in a suitable place. The following are influence the choices of suitable location:

a) Availability of raw materials.

b) Availability of labour.

c) Availability of power.

d) Availability of market

e) Availability of water

f) Availability of Infra structural. [e.g. Transport, Insurance, Telephone, Postal services.]

g) Availability of Govt, rules and regulation.

3. Plant Layout:

It refers to plant layout of arrangement of productive facility in such a way to reduce the cost of production.

i. Process layout

ii. Product layout

iii. Fixed layout

iv. Combination layout.

4. Plant Maintenance:

It refers to the process of up keeping the machine and equipments in good operating conditions.

Type of Plant Maintenance:

i. Routine Maintenance

ii. Breakdown Maintenance

iii. Preventive Maintenance

iv. Productive Maintenances

v. Predictive Maintenance

vi. Maintenance provision.

5. Production planning and controlling (PPC)

Production Planning and Controlling is consists of Planning, Routine, Scheduling, Dispatching and Follow-up function of the process of production. This function are organized in such a way there the performance of machine, movement of materials and operative of labours are direct to the product.

These are the function are mainly used to reduce the cost of production..

Production planning:

It is a process of determining a future production of a particular product.

Routing:

It is a process of determining a best and cheapest route for the production, Route means arrangement of different machine involved in the production.

Scheduling:

These processes are finalizing starting time and completing time for each route.

Dispatching:

It is the process of authorizing the workers to start plan. It refers to the process of issuing necessary job order, tool order, move order, work order, inspection order expect to the work

Follow-up:

It is the process of verifying whether the works going on as per the plan, if not taking necessary action to correcting.

6. Work Study:

It is the productive technique used to investigate all factors, affecting the efficiency economic of human works and to find out bare and needs for improvement.

Work study divided in to two ways

1. Method study

2. **Time Study**

Method study

It is the process of identity the best way of doing the work.

Time Study:

It is the process of fix the standard time for each element of the job.

7. Production System:

It refers the arrangement of various production and operations. Production of a product, requires number of activities, these activities are to be done in a systematic way. If these system is not identify correctly the production, will be affected.

1. Continuous production system
2. Intermittent production system.

8. Dependent Services:

a) Product Standardization

It is the process of fixing standard characteristics for a product.

b) Product Simplification

It is the process of making the product less complicated and simple.

c) Product specialization

It is the process of become a specialist in production of one particular item.

d) Inspection and Quality control

It is a technique of established to control the quality several of the product. Inspection is a process of checking whether the product Processed in specified quality characters.

e) Inventory control

It is a process of determining,

What to stock?

How to stock?

How much to stock?

f) Product Diversification

It is the process of producing different types of products by the same company.

g) Research and Development

Research and Development is normally deals with introduction of new product, modification of new product, simplification of new product, develop model for new product.

h) Employee amenities

The production manager has to take care of the welfare of his worker to provide necessary facilities to the worker.

EVALUATION OF OPERATION MANAGEMENT

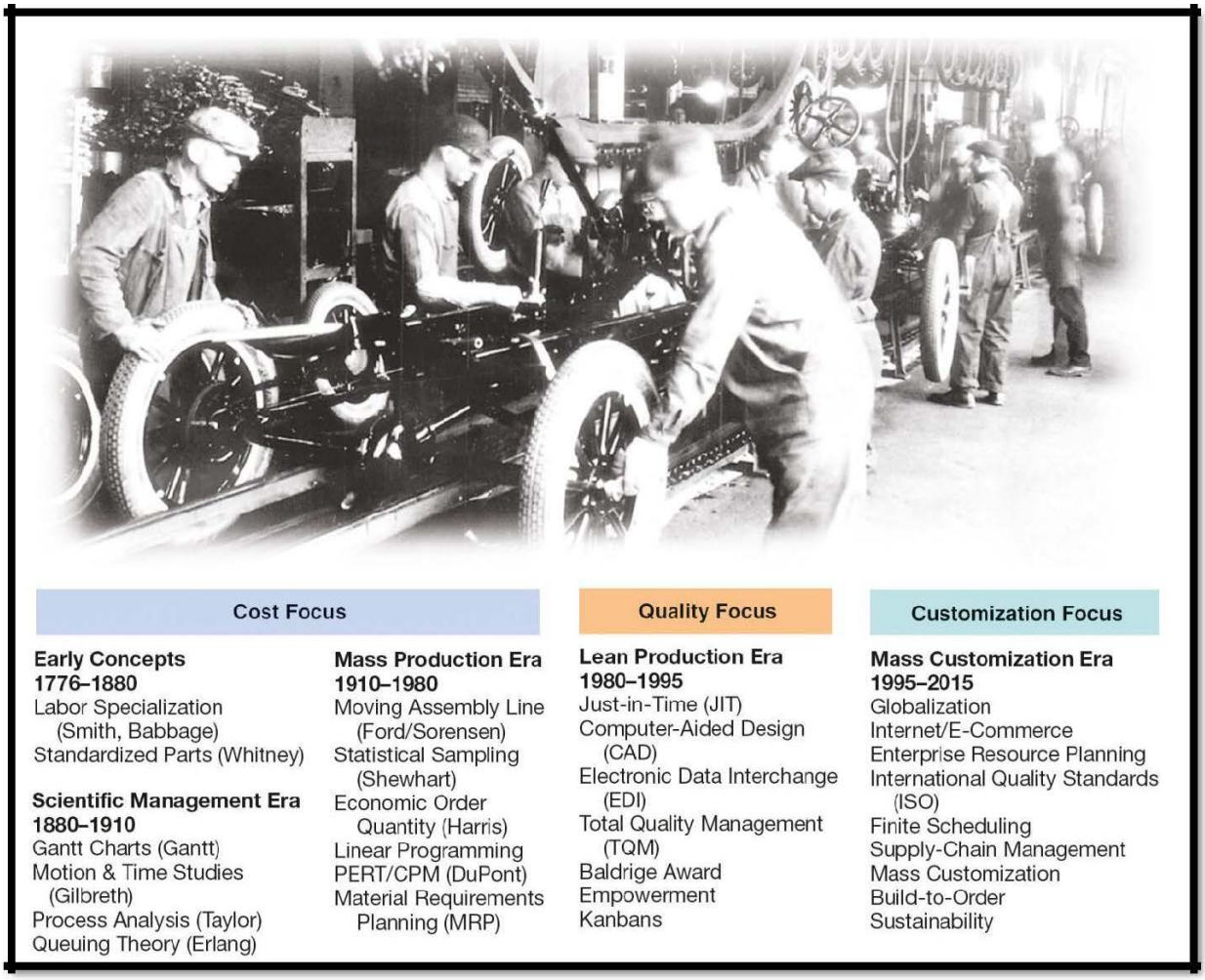
The traditional view of manufacturing management began in eighteenth century when Adam Smith recognised the economic benefits of specialization of labour. He recommended breaking of jobs down into subtasks and recognises workers to specialized tasks in which

they would become highly skilled and efficient. In the early twentieth century, F.W. Taylor implemented Smith‘s theories and developed scientific management. From then till 1930, many techniques were developed prevailing the traditional view. Brief information about the contributions to manufacturing management is shown in the Table 1.1.

Production Management becomes the acceptable term from 1930s to 1950s. As F.W. Taylor‘s works become more widely known, managers developed techniques that focused on economic efficiency in manufacturing. Workers were studied in great detail to eliminate wasteful efforts and achieve greater efficiency.

At the same time, psychologists, socialists and other social scientists began to study people and human behaviour in the working environment. In addition, economists, mathematicians, and computer socialists contributed newer, more sophisticated analytical approaches.

Date	Contribution	Contributor
1776	Specialization of labour in manufacturing	Adam Smith
1799	Interchangeable parts, cost accounting	Eli Whitney & others
1832	Division of labour by skill; assignment of jobs by Skill; basics of time study	Charles Babbage
1900	Scientific management time study and work study Developed; dividing planning and doing of work	Frederick W.Taylor
1900	Motion of study of jobs	Frank B. Gilbreth
1901	Scheduling techniques for employees, machines Jobs in manufacturing	Henry L. Gantt
1915	Economic lot sizes for inventory control	F.W. Harris
1927	Human relations; the Hawthorne studies	Elton Mayo
1931	Statistical inference applied to product quality: quality control charts	W.A. Shewart
1935	Statistical Sampling applied to quality control: inspection sampling plans	H.F.Dodge & H.G.Roming
1940	Operations research applications in world war II	P.M.Blacker & others
1946	Digital Computer	John Mauchlly and J.P.Eckert
1947	Linear Programming	G.B.Dantzig, Williams & others
1950	Mathematical programming, on-linear and stochastic processes	A.Charnes, W.W.Cooper & others
1951	Commercial digital computer: large-scale computations available	Sperry Univac
1960	Organisational behaviour: continued study of people at work	L.Cummings, L.Porter
1970	Integrating operations into overall strategy and policy Computer applications to manufacturing, scheduling, and control, Material Requirement Planning (MRP)	W.Skinner J.Orlicky & G. Wright
1980	Quality and productivity applications from Japan: robotics, CAD-CAM	W.E. Deming & J.Juran



With the 1970s emerge two distinct changes in our views. The most obvious of these, reflected in the new name Operations Management was a shift in the service and manufacturing sectors of the economy. As service sector became more prominent, the change from production 'to operations' emphasized the broadening of our field to service organizations. The second, more suitable change as the beginning of an emphasis on synthesis, rather than just analysis, in management practices.

PRODUCTION SYSTEM

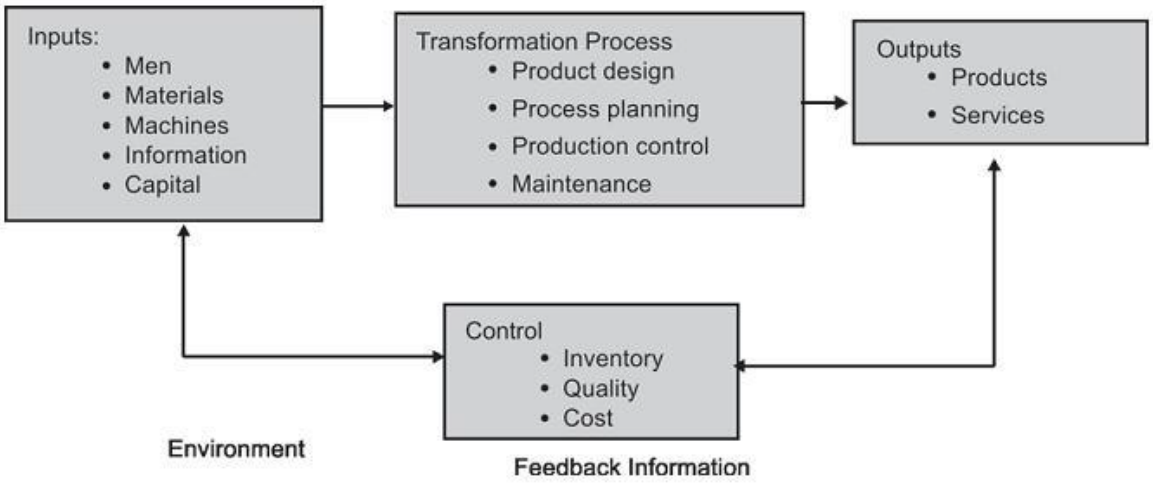
The production system is that part of an organization, which produces products of an organization. It is that activity whereby resources, flowing within a defined system, are combined and transformed in a controlled manner to add value in accordance with the policies communicated by management'.

A simplified production system is shown below:

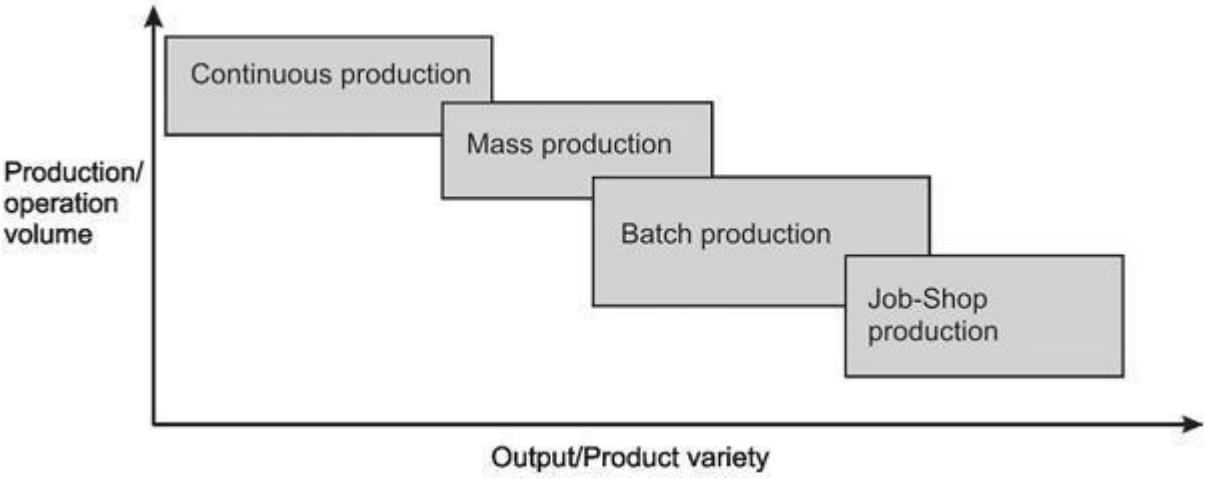
The production system has the following characteristics:

1. Production is an organised activity, so every production system has an objective.
2. The system transforms the various inputs to useful outputs.
3. It does not operate in isolation from the other organisation system.
4. There exists a feedback about the activities, which is essential to control and improve system performance.

PRODUCTION OR CONVERSION PROCESS



CLASSIFICATION OF PRODUCTION SYSTEM



a) Job-Shop Production

Job-shop production are characterised by manufacturing one or few quantity of products designed and produced as per the specification of customers within prefixed time and cost. The distinguishing feature of this is low volume and high variety of products.

A job-shop comprises of general-purpose machines arranged into different departments. Each job demands unique technological requirements, demands processing on machines in a certain sequence.

Job-shop Production is characterized by

1. High variety of products and low volume.
2. Use of general purpose machines and facilities.
3. Highly skilled operators who can take up each job as a challenge because of uniqueness.
4. Large inventory of materials, tools, parts.
5. Detailed planning is essential for sequencing the requirements of each product, capacities for each work centre and order priorities.

Advantages

Following are the advantages of Job-shop Production:

1. Because of general purpose machines and facilities variety of products can be produced.
2. Operators will become more skilled and competent, as each job gives them learning opportunities.
3. Full potential of operators can be utilized.
4. Opportunity exists for Creative methods and innovative ideas.

Limitations

Following are the limitations of Job-shop Production:

1. Higher cost due to frequent set up changes.
2. Higher level of inventory at all levels and hence higher inventory cost.
3. Production planning is complicated.
4. Larger space requirements.

Batch Production

American Production and Inventory Control Society (APICS) defines Batch Production as a form of manufacturing in which the job pass through the functional departments in lots or batches and each lot may have a different routing. It is characterised by the manufacture of limited number of products

produced at regular intervals and stocked awaiting sales.

Batch Production is characterized by

1. Shorter production runs.
2. Plant and machinery are flexible.
3. Plant and machinery set up is used for the production of item in a batch and change of set up is required for processing the next batch.
4. Manufacturing lead-time and cost are lower as compared to job order production.

Advantages

Following are the advantages of Batch Production:

1. Better utilisation of plant and machinery.
2. Promotes functional specialisation.
3. Cost per unit is lower as compared to job order production.
4. Lower investment in plant and machinery.
5. Flexibility to accommodate and process number of products.
6. Job satisfaction exists for operators.

Limitations

Following are the limitations of Batch Production:

1. Material handling is complex because of irregular and longer flows.
2. Production planning and control is complex.
3. Work in process inventory is higher compared to continuous production.
4. Higher set up costs due to frequent changes in set up.

Mass Production

Manufacture of discrete parts or assemblies using a continuous process are called Mass Production. This production system is justified by very large volume of production. The machines are arranged in a line or product layout. Product and process standardization exists and all outputs follow the same path.

Mass Production is characterized by

1. Standardization of product and process sequence.
2. Dedicated special purpose machines having higher production capacities and output rates.
3. Large volume of products.
4. Shorter cycle time of production.
5. Lower in process inventory.
6. Perfectly balanced production lines.
7. Flow of materials, components and parts is continuous and without any back tracking.

8. Production planning and control is easy.

9. Material handling can be completely automatic.

Advantages

Following are the advantages of Mass Production:

1. Higher rate of production with reduced cycle time.

2. Higher capacity utilisation due to line balancing.

3. Less skilled operators are required.

4. Low process inventory.

5. Manufacturing cost per unit is low.

Limitations

Following are the limitations of Mass Production:

1. Breakdown of one machine will stop an entire production line.

2. Line layout needs major change with the changes in the product design.

3. High investment in production facilities.

4. The cycle time is determined by the slowest operation.

Continuous Production

Production facilities are arranged as per the sequence of production operations from the first operations to the finished product. The items are made to flow through the sequence of operations through material handling devices such as conveyors, transfer devices, etc.

Continuous Production is characterized by

1. Dedicated plant and equipment with zero flexibility.

2. Material handling is fully automated.

3. Process follows a predetermined sequence of operations.

4. Component materials cannot be readily identified with final product.

5. Planning and scheduling is a routine action.

Advantages

Following are the advantages of Continuous Production:

1. Standardisation of product and process sequence.

2. Higher rate of production with reduced cycle time.

3. Higher capacity utilisation due to line balancing.

4. Manpower is not required for material handling as it is completely automatic.

5. Person with limited skills can be used on the production line.

6. Unit cost is lower due to high volume of production.

Limitations

Following are the limitations of Continuous Production:

1. Flexibility to accommodate and process number of products does not exist.
2. Very high investment for setting flow lines.
3. Product differentiation is limited.

DIFFERENCE BETWEEN IPS AND CPS

S.NO	CHARACTERISTICS	IPS	CPS
1	**Product**	Any product	Specific product
2	**Production volume**	Small	Large
3	**Machine**	General purpose	Specific purpose
4	**Machine setting**	Frequently machine setting	Less machine setting
5	**Production policy**	Production to order	Production to stock
6	**Types of labours**	Skilled	Semi and unskilled
7	**Maintenance system**	Breakdown maintenance	Preventive maintenance
8	**Material handling equipment**	Variable path	Fixed path
9	**Production line balancing**	Not necessary	More important
10	**PPC**	Tough difficult	One time job
11	**Store keeping**	Large storage place	Limited storage
12	**Investment**	Less capital investment	Huge Capital investment
13	**Variety of product**	Is possible	Is not possible
14	**Adjustment of product**	Is possible	Is possible
15	**Quality of the product**	Difficult	Produce quality product

MODERN PRODUCTION AND OPERATIONS FUNCTION

The production management of today presents certain characteristics which make it look totally different from what it was during the past. Specifically, today production system is characterized by basically four features:

1. Manufacturing as Competitive Advantage:

In the past, production was considered to be like any other function in an organization. When demand was high and production capacities were inadequate , the concern was to somehow muster all inputs and use them to produce goods which would be grabbed by the potential market. But today scenario is contrasting. Plants have excess capacities, competition is mounting and firms look and gain competitive advantage to survive and succeed.

Interestingly system offers vast scope to gain a competitive edge and firms intend to exploit the potential. Total Quality Management (TQM) , Time-Based Competition, Business Process Reengineering (BPRE) , Just in Time (JIT), Focused Factory (Revised) , Flexible Manufacturing Systems(FMS) , Computer Integrated Manufacturing (CIM) and The Virtual Corporation are but only some techniques which the companies are employing to gain competitive advantage.

CHARACTERISTICS OF SYSTEMS TO PRODUCE PRODUCTS VERSUS SYSTEM TO PRODUCE SERVICES.

Products	Service
Tangible	Intangible and perishable; consumed in the Process of their production
Can be produced to inventory for-off the Shelf availability	Availability achieved by keeping the productive System open for services.
Minimal contact with ultimate consumer	High contact with clients or customers.
Complex and interrelated processing	Simple processing
Demand on system variable on weekly, monthly an seasonal basis	Demand commonly variable on hourly, daily and weekly basis
Markets served by production system are regional, national and international	Markets served by production system are usually local.
Large units that can take advantage of economies of scale	Relatively small units to serve local markets
Location of system is in relation to regional, national and international markets.	Location, dependent on location of local customers, clients and users.

2. Service Orientation :

As stated earlier , the service sector is gaining greater relevance these days. The production system, therefore , needs to be organized , keeping in mind that peculiar requirements of the service component .As was shown in the above table, the entire manufacturing needs to be geared to serve

(i) intangible and perishable nature of the services

(ii) constant interaction with clients or customers,

(iii) small volumes of production to serve local markets and

(iv) need to locate facilities to serve local markets.

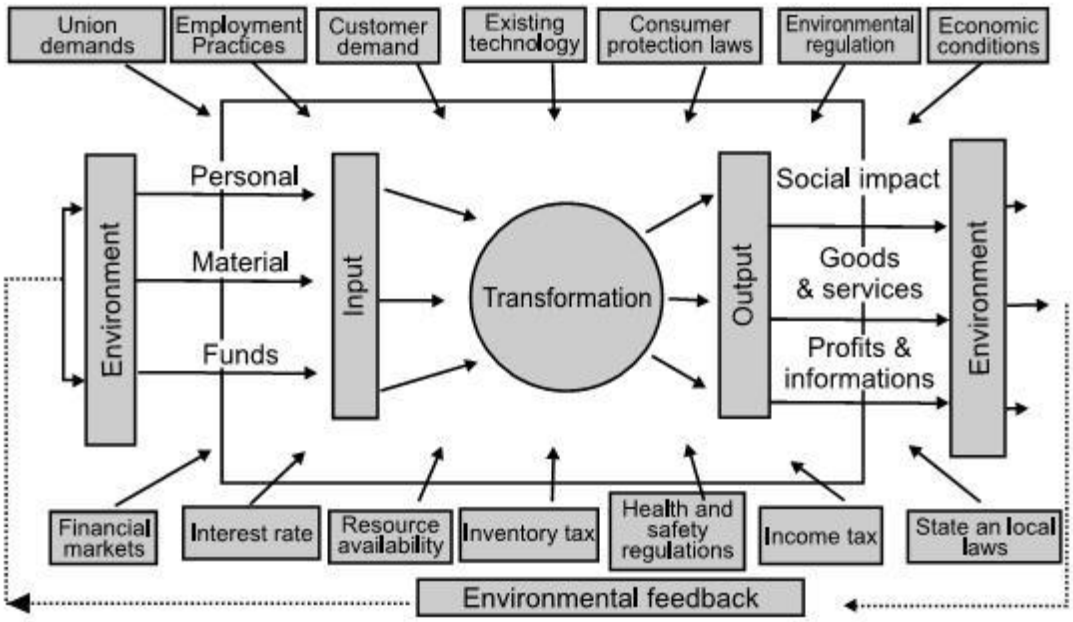
There is increased presence of professionals on the production side, instead of technicians and engineers.

3. Disappearance of Smokestacks:

Commencing from the Industrial Revolution till the middle of the 20th century, production system was dominated by smokestacks .These smokestacks (the term used by Alvin Toffler in his book Power Shift) represented industrial establishments which ejected thick smoke, polluting the environment around. Smokestacks not only disgorged reek, they produced nauseating smell, generated dust, created sound and in general , were resembling ghosts .Not that they have become extinct, but, they are disappearing gradually. Protective labor legislation, environmental movement and gradual emergence of knowledge based organizations, have brought total transformation in the production system. Today's factories are aesthetically designed and built , environment friendly- in fact, they are, a home away from home. Going to factory everyday is no more an excruciating experience , it is like enjoying work with a healthy perspective. A visit to ABB, L&T or Smith-Kline and Beecham should convince the reader about the transformation, that has taken place in the wealth creation system.

THE ENVIRONMENT OF OPERATIONS

One of the most encompassing influences on productivity is the environment in which organization operates. The social impact of an organization is a reflection of the values held by top management.



It evolves from the religious and cultural norms of society, from childhood training, education, and reflection on the purpose of life and the value of one's self and of others. The

preferred values of the society reflect purpose, integrity and a respect for the life and humanity of others. Every facet of our economic and social environment regulated and controlled by law designed to protect general public.

The above figure illustrates some of the laws impact on the productivity of the firms.

FUNCTIONS OF PRODUCTION MANAGEMENT DEPARTMENT

The functions of Production Management depend upon the size of the firm. In small firms the production Manager may have to look after production planning and control along with Personnel, Marketing, Finance and Purchase functions. In medium sized firms, there may be separate managers for Personnel, marketing and Finance functions. But the production planning and control and Purchase and stores may be under the control of Production management department. In large sized firms the activities of Production Management is confined to the management of production activities only. As such, there are no hard and fast rule or guidelines to specify the function of Production Management, but in the academic interest we can mention some of the functions, which are looked after by the Production Management department. They are:

(i) Materials: The selection of materials for the product. Production manager must have sound Knowledge of materials and their properties, so that he can select appropriate materials for his product. Research on materials is necessary to find alternatives to satisfy the changing needs of the design in the product and availability of material resumes.

(ii) Methods: Finding the best method for the process, to search for the methods to suit the available resources, identifying the sequence of process are some of the activities of Production Management.

(iii) Machines and Equipment: Selection of suitable machinery for the process desired, designing the maintenance policy and design of layout of machines are taken care of by the Production Management department.

(iv) Estimating: To fix up the Production targets and delivery dates and to keep the production costs

at minimum, production management department does a thorough estimation of Production times and production costs. In competitive situation this will help the management to decide what should be done in arresting the costs at desired level.

(v) Loading and Scheduling: The Production Management department has to draw the time table for various production activities, specifying when to start and when to finish the process required. It also has to draw the timings of materials movement and plan the activities of

manpower. The scheduling is to be done keeping in mind the loads on hand and capacities of facilities available.

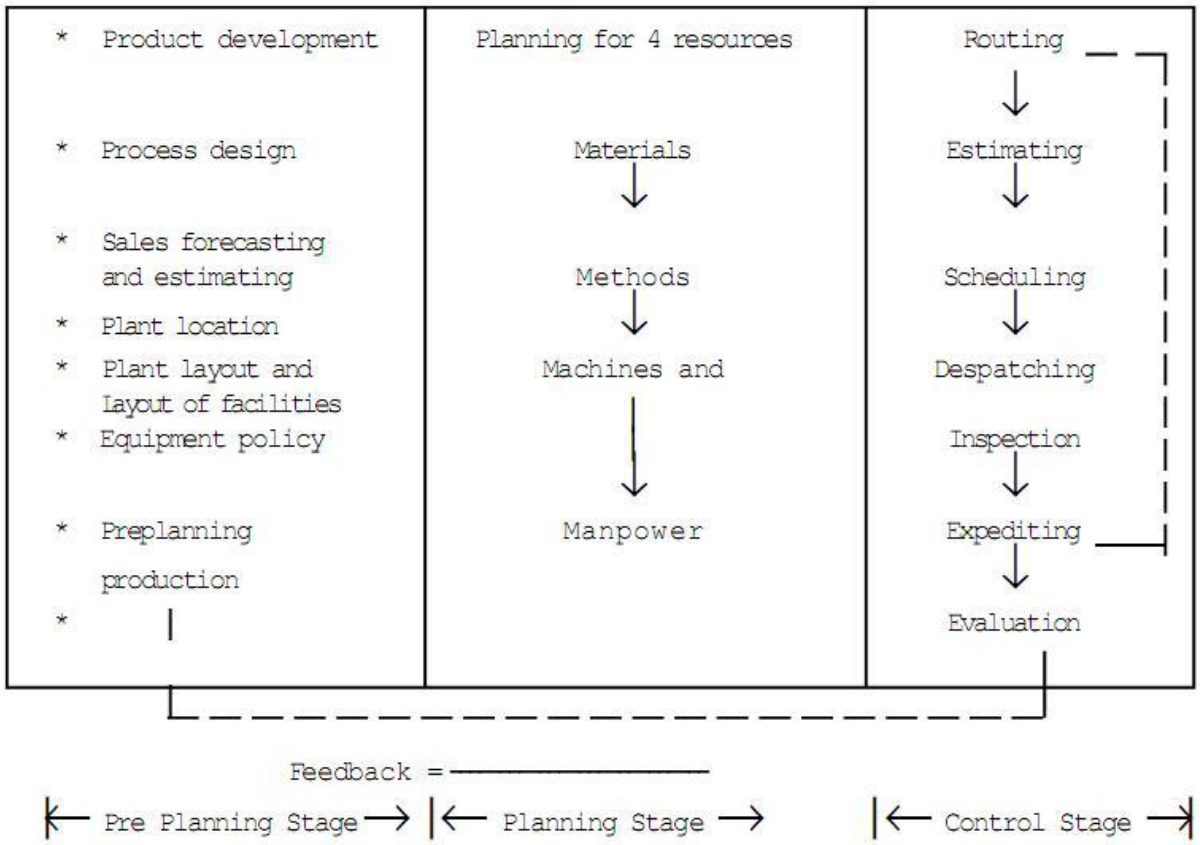
(vi) Routing: This is the most important function of Production Management department. The Routing consists of fixing the flow lines for various raw materials, components etc., from the stores to the packing of finished product, so that all concerned knows what exactly is happening on the shop floor.

(vii) Dispatching: The Production Management department has to prepare various documents such as Job Cards, Route sheets, Move Cards, Inspection Cards for each and every component of the product. These are prepared in a set of five copies. These documents are to be released from Production Management department to give green signal for starting the production. The activities of the shop floor will follow the instructions given in these documents. Activity of releasing the document is known as dispatching.

(viii) Expediting or Follow up: Once the documents are dispatched, the management wants to know whether the activities are being carried out as per the plans or not. Expediting engineers go round the production floor along with the plans, compare the actual with the plan and feed back the progress of

(ix) Inspection: Here inspection is generally concerned with the inspection activities during production, but a separate quality control department does the quality inspection, which is not under the control of Production Management. This is true because, if the quality inspection is given to production Management, then there is a chance of qualifying the defective products also. For example Teaching and examining of students is given to the same person, then there is a possibility of passing all the students in the first grade. To avoid this situation an external person does correction of answer scripts, so that the quality of answers is correctly judged.

PRODUCTION FUNCTION



(x) Evaluation: The Production department must evaluate itself and its contribution in fulfilling the corporate objectives and the departmental objectives. This is necessary for setting up the standards for future. What ever may be the size of the firm; Production management department alone must do Routing, Scheduling, Loading, Dispatching and expediting. This is because this department knows very well regarding materials, Methods, and available resources etc.

If the firms are small, all the above-mentioned functions (i to x) are to be carried out by Production Management Department. In medium sized firms in addition to Routing, Scheduling and Loading,

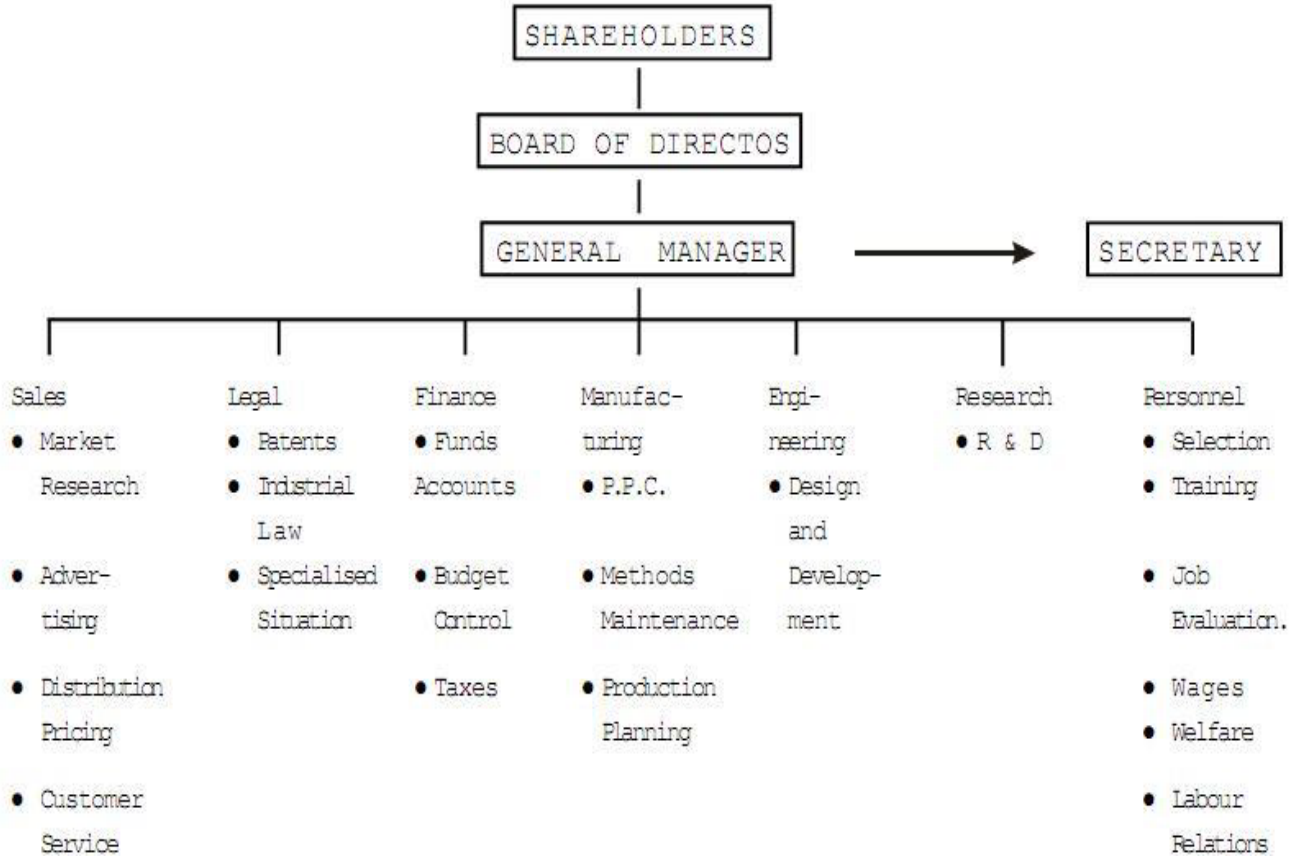
Dispatching and expediting, some more functions like Methods, Machines may be under the control of Production Management Department. In large firms, there will be Separate departments for Methods, Machines, Materials and others but routing, loading and scheduling are the sole functions of Production Management. All the above ten functions are categorized in three stage, that is Preplanning, Planning and control stages as shown in figure.1.2.the work to the management. This will help the management to evaluate the plans.

PRODUCTION MANAGEMENT DEPARTMENT IN THE ORGANISATION

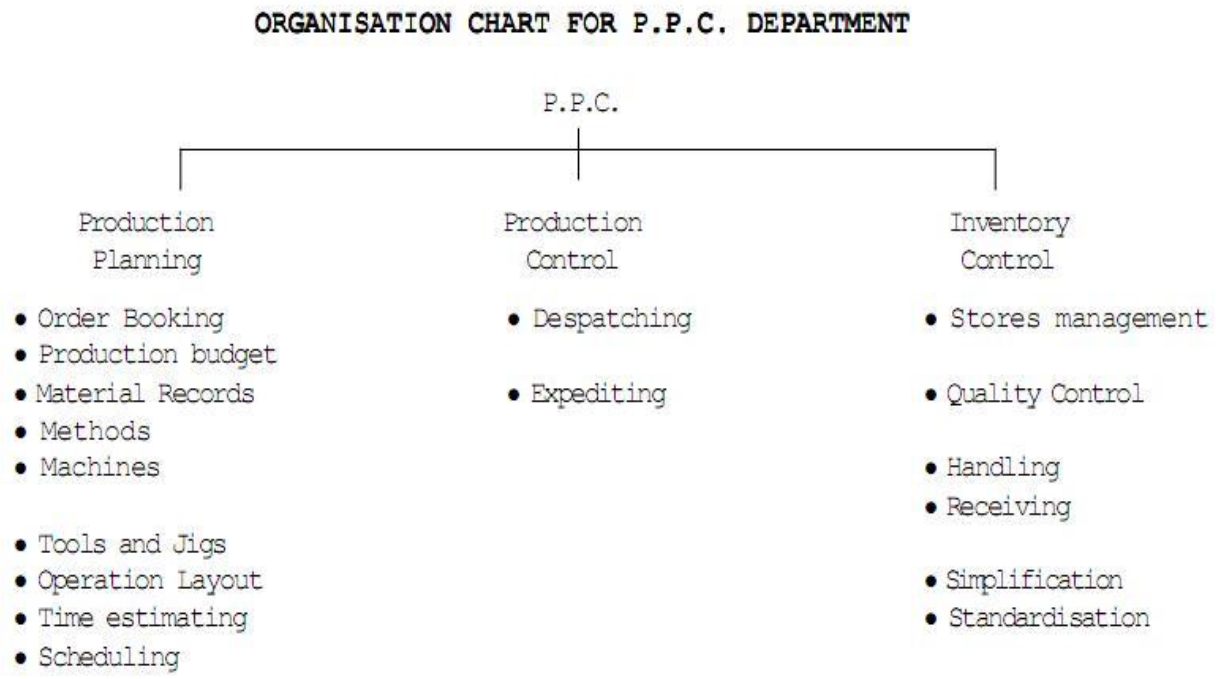
Production is the center of all activities of an organization. This is to say all the activities of an organization, such as: Finance, Personnel, Marketing...etc., are exists in an organization because of production activity. Hence the position of Production Management in an organization is very important.

Whether it should be a line function or Staff function, more or less depends upon the corporate management policy. In small organization, Production Management is whole and sole of it. In industries, generally it is advisable to have Production as line function, because, the decision taken by the line manager and the advices given by the Staff personnel will be based on the Production activities. The Production Manager, directly report to General Manager, who in turn report to the Board. The figure 1.3 shows a typical organization structure showing the position of Production Management. Figure 1.4 shows an organization chart for Production and Operations Management department.

TYPICAL ORGANIZATION CHART FOR AN ORGANIZATION



AN ORGANIZATION CHART FOR PRODUCTION MANAGEMENT DEPARTMENT



OPERATIONS STRATEGY

Operations strategy is concerned with setting broad policies and plans for using the resources of the firm to best support the firm‘s long-term competitive strategy.

Operations Management Decisions

1. Strategic (long-term) decisions

Well the relevant points over here are-

How will we make the product?

Where do we locate the facility or facilities?

How much capacity do we need?

When should we add more capacity?

2. Tactical (intermediate-term) decisions

Well, the relevant points over here are-

How many workers do we need?

When do we need them?

Should we work overtime or put on a second shift?

When should we have material delivered?

Should we have material delivered?

Should we have a finished goods inventory?

3. Operational Planning & Control (short-term) decisions

Well, the relevant points over here are-

What jobs do we work on today or this week?

Who do we assign to what tasks?

What jobs have priority?

An operations strategy can be viewed as part of a planning process that coordinates operational goals with those of the larger organization. Since the goals of the larger organization change over time, the operations strategy must be designed to anticipate future needs.

An operations strategy involves decisions that relate to:

a) Process design includes:

1) the selection of appropriate technology;

2) sizing the process over time;

3) the role of inventory in the process; and,

4) locating the process.

b) Infrastructure decisions involve:

1) the logic associated with the planning and control systems;

2) quality assurance and control approaches;

3) work payment structures; and,

4) the organization of the operations function.

Strategy is a common vision that unites an organization, provides consistency in decisions, and keeps the organization moving in the right direction. Strategy formulation consists of four basic steps:

Defining a primary task: The primary task is usually expressed in a firm‘s mission statement. The mission may be accompanied by a vision statement that describes what the organization sees itself becoming.

Assessing core competencies: Core competency is what a firm does better than anyone else, its distinctive competence. Based on experience, knowledge, and know-how, core competencies represent sustainable competitive advantages. For this reason, products and technologies are seldom core competencies.

The advantage they provide is short-lived, and other companies can readily purchase, emulate or improve on them. Core competencies are more likely to be processes, a company‘s ability to do certain things better than a competitor. Thus, while a particular product is not a core

competence, the process of developing new products is. Core competencies are not static. They should be nurtured, enhanced, and developed over time. Close contact with the customer is essential to ensuring that a competence does not become obsolete.

Determining order winners and order qualifiers:

A firm is in trouble if the things it does best are not important to the customer. That‘s why it‘s essential to look toward customers to determine what influences their purchase decision. Order qualifiers are the characteristics of a product or service that qualify it to be considered for purchase by a customer. An order winner is the characteristic of a product or service that wins orders in the marketplace-the final factor in the purchasing decision.

THE STRATEGIC ROLE OF OPERATIONS

Primary goals of the organisations are related market opportunities. Economy and efficiency of conversion operations are the secondary goals, which will be predominant with the study and practice of operations management.

A STRATEGIC PERSPECTIVE

In figure 1.1 provides the basic downward flow of strategy influence leading to managing conversion operations and results. The general thrust of the process is guided by competitive and market conditions in the industry, which provide the basis for determining the organization‘s strategy. Where is the industry now, and where it will be in the future?

What are the existing and potential markets?

What market gaps exist, and what competencies do we have for filling them?

A careful analysis of market segments and the ability of our competitors and ourselves to meet the needs of these segments will determine the best direction for focusing an organization‘s efforts.

After assessing the potential within an industry, an overall organizational strategy must be developed, including some basic choices of the primary basis for competing. In doing so, priorities are established among the following four characteristics:

• Quality (product performance).

• Cost efficiency (low product price).

• Dependability (reliable, timely delivery of orders to customers).

• Flexibility (responding rapidly with new products or changes in volume).

In recent years, most organizations cannot be best on all these dimensions and, by trying to do so, they end up doing nothing well. Furthermore, when a competency exists in one of

these areas, an attempt to switch to a different one can lead to a downfall in effectiveness (meeting the primary objectives).

Time is emerging as a critical dimension of competition in both manufacturing and service industries. In any industry the firm with the fastest response to customer demands has the potential to achieve an overwhelming market advantage. In an era of time-based competition, a firm's competitive advantage is defined by the total time required to produce a product or service. Firms able to respond quickly have reported growth rates over three times the industry average and double the profitability. Thus the pay-off for quick response is market dominance. These basic strategic choices set the tone for the shape and content of the operations functions.

OPERATIONS OBJECTIVES

The overall objective of the operations subsystem is to provide conversion capabilities for meeting the organization‘s goals and strategy. The sub-goals of the operations sub system, must specify the following:

1. Product/service characteristics.

2. Process characteristics.

3. Product/service quality.

4. Efficiency

- Effective employee relations and cost control of labour.
- Cost control of material.
- Cost control in facility utilization.

5. Customer service (schedule)

- Producing quantities to meet expected demand.
- Meeting the required delivery date for goods or services.

6. Adaptability for future survival.

The priorities among these operations ‘sub-goals and their relative emphases should be direct reflections of the organization‘s mission. Relating these six operations sub-goals to the broader strategic choices above, it is clear that quality, efficiency, and dependability (customer service) are reflected in the sub goals.

Flexibility encompasses adaptability but also relates to product/service and process characteristics: Once choices about product and process are made, boundaries for meeting the other operations objectives are set.

OPERATIONS ALTERNATIVES AND TRADEOFFS

The operations sub-goals can be attained through the decisions that are made in the various operations areas. Each decision involves important tradeoffs between choices about product and process versus choices about quality, efficiency, schedule and adaptability.

Once a decision is made, it leads to many choices. Where should facilities be located? How large should they be? What degree of automation should be used? How skilled must labour be to operate the automated equipment? Will the product be produced on site? How do these decisions impact quality, efficiency, schedule (customer service), and adaptability? Are we prepared for changes in product or service, or do these decisions lock in our operations? These are examples of the tough, crucial tradeoffs that are at the heart of understanding the choices that must be made when planning strategically and tactically.

STRATEGIC PLANNING

Strategic planning is the process of thinking through the current mission of the organization and the current environmental conditions facing it, then setting forth a guide for tomorrow‘s decisions and results. Strategic planning is built on fundamental concepts: those current decisions are based on future conditions and results.

Strategic Planning for Production and Operations

In the production or operations function, strategic planning is the broad, overall planning that precedes the more detailed operational planning. Executives who head the production and operations function are actively involved in strategic planning, developing plans that are consistent with the firm‘s overall strategies as well as such functions as marketing, finance accounting and engineering. Production and operations strategic plans are the basis for

(1) operational planning of facilities (design) and

(2) operational planning for the use of these facilities.

Strategic Planning Approaches for Production/Operations

Henry Mintzberg suggests three contrasting modes of strategic planning: the entrepreneurial, the adaptive, and the planning modes. In the entrepreneurial mode, one strong, bold leader takes planning action on behalf of the production/operations function. In the adaptive mode, a manager‘s plan is formulated in a series of small, disjointed steps in reaction to a disjointed environment. The planning model uses planning essentials combined with the logical analysis of management science.

There are many approaches to strategic planning. The key point is that operations strategies must be consistent with the overall strategies of the firm. Operations typically utilize the

overall corporate approach to strategic planning, with special modifications and a focus upon operations issues and opportunities. One general approach to strategic planning is a forced choice model given by Adam and Ebert.

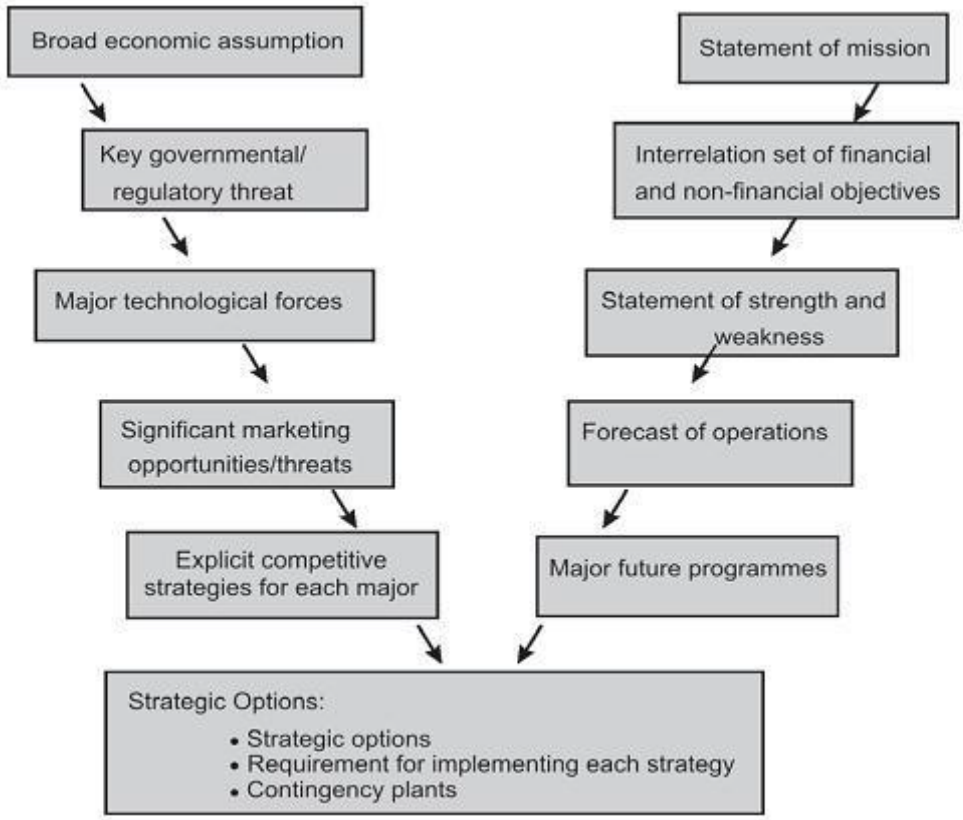
Strategic Planning—Forced Choice Model

One of many planning models that have been used in strategic planning is a forced choice model, shown in Figure 1.6 (on page 16). In-group sessions or individually, analysts assess environmental considerations together with the organization‘s current production/operations position, thus forcing management to develop strategic options for operations.

A Strategic Planning Operations Model

Professor Chris A. Voss of the London Business School, England, has set forth a framework for strategy and policy development in manufacturing. Concept is that manufacturing strategy tries to link the policy decisions associated with operations to the marketplace, the environment, and the company‘s overall goals.

A FORCED CHOICE MODEL OF STRATEGIC PLANNING FOR OPERATIONS



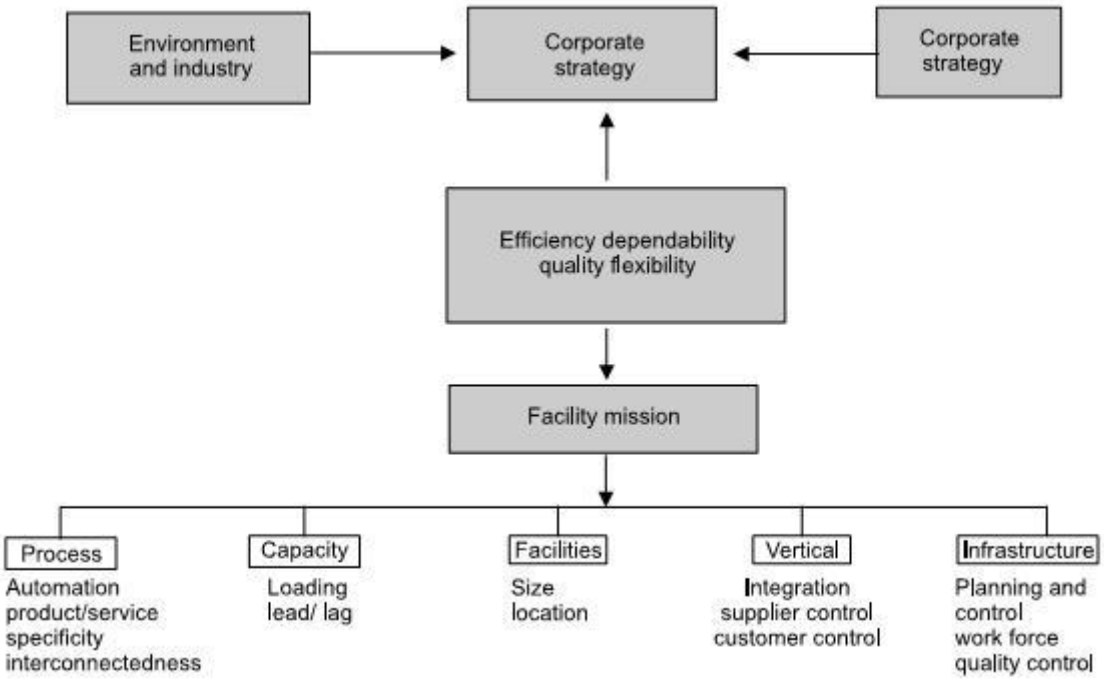
One feature of this approach that is crucial to competitiveness is market-based view of strategic planning. It suggests that any strategic business unit of a company operates in the context of its corporate resources, the general and competitive industry environment, and the

specific corporate goals of the company. In any area in which the company chooses to compete is a set of specific market-based criteria for success. A low-cost, high productivity operation makes efficiency possible. Minimum use of scarce resources while sustaining high outputs is the key to productivity. Effectiveness is how well a company is able to meet specific criteria such as delivery schedules and technical capability.

A simplified framework for examining operations strategy is shown in above Figure.

Quality is the degree to which the product or services meets customer and organisation expectations. Quality reflects the goodness‘ of the product or services to the customer. Flexibility is the adoptability, the capability to change as business conditions change.

OPERATIONS STRATEGY FRAMEWORK



INTERNATIONAL DIMENSIONS OF PRODUCTIVITY

Industrialized nations are developing two strategies to remain competitive in the business.

1. Moving to a new and more advanced products, and
2. Employing better and more flexible system.

NEW PRODUCTS

High Volume Products like steel, textiles, etc. with constitute an industrial base are not secure. Nations such as Japan, France and West Germany are shifting their industrial base

towards products and processes that make better use of their research capabilities and skilled workers. Their future lies in microelectronics, precision manufactured castings, specialty steels, custom fabrics, fiber optics, lasers, etc.

TREND TOWARDS MORE FLEXIBLE SYSTEMS

The production runs of these higher valued specialty items and custom designed products are often much shorter than for traditional mass produced goods. But the non-productive time (downtime) required to set up equipment for producing different options, new models and new products are very costly. So production facilities must be designed with the utmost flexibility to accommodate change overs in rapid fashion. This is where computers, robotics come into play. German executives understand the need for a strong technological focus and the dangers of hierarchical bureaucracies and paper profits. Studies reveal that over 50 per cent of Germany‘s large manufacturing firms are managed by Ph.D.‘s with technical backgrounds.

In recent years, the managerial techniques and productivity methods in Japanese firms have attracted worldwide attention. The following are some of the characteristics of the Japanese firm as compared with the American firms.

1. Corporate objectives: Employees and customers are given priority over shareholders. Honesty in business is important.

2. Time horizon: Long-term viability is more important than short-term profits.

3. Production systems: Automated systems with extensive use of microprocessors and robotics. Quality is paramount, and things happen on schedule.

4. Employment relations: Long-term employment of loyal workers. Unions cooperate to benefit total firm. Politeness and harmony are emphasized.

5. Materials: Resources are limited. Space is used efficiently and inventories are kept to a bare minimum.

6. Financing: More use is made of debt capital and less of equity capital.

7. Training: Employees are thoroughly trained and rotated to learn a variety of skills.

8. Worker participation: Employees are thoroughly trained and rotated to learn a variety of productivity improvements via suggestions, quality circles and consultation with supervisors.

UNIT II FORECASTING, CAPACITY AND FACILITY DESIGN

Demand Forecasting – Need, Types, Objectives and Steps. Overview of Qualitative and Quantitative methods. Capacity Planning – Long range, Types, Developing capacity alternatives. Overview of sales and operations planning. Overview of MRP, MRP II and ERP. Facility Location – Theories, Steps in Selection, Location Models. Facility Layout Principles, Types, Planning tools and techniques.

INTRODUCTION

Forecasts are essential for the smooth operations of business organizations. They provide information that can assist managers in guiding future activities toward organizational goals.

FORECASTING OBJECTIVES AND USES

Forecasting is the art and science of predicting future events. Forecasts are required throughout an organization and at all levels of decision making in order to plan for the future and make effective decisions.

Forecasts are estimates of the occurrence, timing, or magnitude of uncertain future events. Forecasts are essential for the smooth operations of business organizations. They provide information that can assist managers in guiding future activities toward organizational goals.

Operations managers are primarily concerned with forecasts of demand-which are often made by (or in conjunction with) marketing. However, managers also use forecasts to estimate raw material prices, plan for appropriate levels of personnel, help decide how much inventory to carry, and a host of other activities. This results in better use of capacity, more responsive service to customers, and improved profitability.

FORECASTING DECISION VARIABLES

Forecasting activities are a function of

(1) the type of forecast (e.g., demand, technological),

(2) the time horizon (short, medium, or long range),

(3) the database available, and

(4) the methodology employed (qualitative or quantitative).

Forecasts of demand are based primarily on non-random trends and relationships, with an allowance for random components. Forecasts for groups of products tend to be more accurate than those for single products, and short-term forecasts are more accurate than long-term forecasts (greater than five years). Quantification also enhances the objectivity and precision of a forecast.

TYPES OF FORECASTS

In general, a contemporary business organization employs three distinct types of forecasts. These are given under:

1. Economic forecasts: Economic forecasts address the business cycle by predicting inflation rates, money supplies, housing starts, and other planning indicators.
2. Technological forecasts: Technological forecasts are concerned with rates of technological progress, which can result in the birth of exciting new products, requiring new plants and equipment.
3. Demand forecasts: Demand forecasts are projections of demand for a company‘s products or services.

These forecasts, also called sales forecasts, drive a company‘s production, capacity, and scheduling systems and serve as inputs to financial, marketing, and personnel planning.

STRATEGIC IMPORTANCE OF FORECASTING:

Forecasting plays a very important role in the following areas:

A) HUMAN RESOURCE MANAGEMENT - hiring, training and laying-off workers all depend on anticipated demand.)

B) CAPACITY PLANNING - when capacity is inadequate, the resulting shortages can mean undependable delivery, loss of customers, and loss of market share.)

C) SUPPLY CHAIN MANAGEMENT - good supplier relations and the ensuing price advantages for materials and parts depend on accurate forecasts.)

STEPS IN FORECASTING

There are eight steps to a forecasting system. These are:

1. Determine the use of the forecast – (What objectives are we trying to achieve?)
2. Select the items that are to be forecasted
3. Determine the time horizon of the forecast – (Is it short, medium, or long – range?)
4. Select the forecasting model
5. Gather the data needed to make the forecast
6. Validate the forecasting model
7. Make the forecast
8. Implement the results

FORECASTING METHODS:

There are numerous approaches to forecasting depending on the need of the decision maker. Broadly speaking, these can be categorized in two ways:

i) Quantitative forecasting

ii) Qualitative forecasting

The various **Qualitative Methods** in vogue are as follows:

1. Jury of executive opinion – This method takes the opinions of a small group of high-level managers, often in combination with statistical models, and results in a group estimate of demand.

2. Sales force composite – In this approach, each salespeople estimates what sales will be in his or her region. These forecasts are then reviewed to ensure they are realistic, then combined at the district and national levels to reach an overall forecast.

3. Delphi method – This is an iterative group process. There are three different types of participants in the Delphi process: decision makers, staff personnel, and respondents. The decision makers usually consist of a group of five to ten experts who will be making the actual forecast. The staff personnel assist the decision makers by preparing, distributing, collecting, and summarizing a series of questionnaires and survey results. The respondents are a group of people whose judgments are valued and are being sought. This group provides inputs to the decision makers before the forecast is made.

4. Consumer market survey – This method take input from customers or potential customers regarding their future purchasing plans. It can help not only in preparing a forecast but also in improving product design and planning for new products.

5. Naïve approach – It assumes that demand in the next period is the same as demand in the most recent period. In other words, if sales of a product, say, Reliance WLL phones, were 100 units in January, we can forecast that February‘s sales will also be 100 phones. Does this make any sense? It turns out that for some product lines, selecting this naïve approach is a cost-effective and efficient forecasting model.

QUANTITATIVE METHODS

The chief Quantitative methods are:

1. Moving averages
2. Exponential smoothing
3. Time series models
4. Trend projection

The **time series models** of forecasting predict on the basis of the assumption that the future is a function of the past. In other words, they look at what has happened over a period of time and use a series of past data to make a forecast. If we are predicting weekly sales of washing machine, we use the past weekly sales for washing machine in making the forecast.

A **causal model** incorporates into the model the variables or relationships that might influence the quantity being forecast. A causal model for washing machine sales might include relationships such as new housing, advertising budget, and competitors‘ prices. Moving over to a structured approach to forecasting, let me introduce the basic steps involved in this process:-

Time Series Forecasting A time series is based on a sequence of evenly spaced (weekly, monthly, quarterly, and so on) data points. Forecasting time series data implies that future values are predicted only from past values and that other variables, no matter how potentially valuable, are ignored.

Decomposition of a Time Series

There are four main ways of decomposing the time series:

- Trend
- Seasonality
- Cycles
- Random variations

Two general forms of time series models are used in statistics. The most widely used is a multiplicative model, which assumes that demand is the product of the four components:

Demand = T x Rx Cx S

where

T - denotes Trend ,

S - denotes Season

C - denotes Cycles

R - denotes random variables

An additive model provides an estimate by adding the components together. It is stated as:

Demand = T + S + C + R

MOVING AVERAGE

A centered moving average (MA) is obtained by summing and averaging the values from a given number of periods repetitively, each time deleting the oldest value and adding a new

value. Moving averages can smooth out fluctuations in any data, while preserving the general pattern of the data (longer averages result in more smoothing). However, they do not yield a forecasting equation, nor do they generate values for the ends of the data series.

$$MA = \frac{\Sigma X}{\text{Number of Period}}$$

A weighted moving average (MAw) allows some values to be emphasized by varying the weights assigned to each component of the average. Weights can be either percentages or a real number.

$$MAwt = \frac{\Sigma(Wt)\ X}{\Sigma Wt}$$

2. HAND FITTING

A hand fit or freehand curve is simply a plot of a representative line that (subjectively) seems to best fit the data points. For linear data, the forecasting equation will be of the form:

$$Yc = a + b\ (X)\ \text{(signature)}$$

where

Yc is the trend value, a is the intercept (where line crosses the vertical axis),

b is the slope (the rise, Δ y, divided by the run, Δ x), and

X is the time value (years, quarters, etc.).

The signature‖ identifies the point in time when X = 0, as well as the X and Y units.

3. LEAST SQUARES

Least squares are a mathematical technique of fitting a trend to data points. The resulting line of best fit has the following properties:

(1) the summation of all vertical deviations about it is zero,

(2) the summation of all vertical deviations squared is a minimum, and

(3) the line goes through the means X and Y. For linear equations, the line of best fit is found by the simultaneous solution for a and b of the following two normal equations:

$$\mathbf{\Sigma Y = na + b\ \Sigma\ X}$$

$$\mathbf{\Sigma\ XY = a\Sigma X + b\Sigma X}$$

The above equations can be used in the form shown above and are used in that form for regression. However, with time series, the data can also be coded so that $\Sigma = 0$ X . Two terms then dropout, and the equations are simplified to:

$$\Sigma Y = na$$

$$a = \frac{\Sigma Y}{n}$$

$$\Sigma XY = b\Sigma X^2$$

$$b = \frac{\Sigma XY}{X^2}$$

To code the time series data, designate the center of the time span as X = 0 and let each successive period be ±1 more unit away. (For an even number of periods, use values of ±0.5, 1.5, 2.5, etc.).

Seasonal indices:

A seasonal index (SI) is a ratio that relates a recurring seasonal variation to the corresponding trend value at the given time. In the ratio-to-moving average method of calculation monthly (or quarterly) data are typically used to compute a 12-month (or 4-quarter) moving average. (This dampens out all seasonal fluctuations.) Actual monthly (or quarterly) values are then divided by the moving average value centered on the actual month. In the ratio-to-trend method, the actual values are divided by the trend value centered on the actual month. The ratios obtained for several of the same months (or quarters) are then averaged to obtain the seasonal index values. The indexes can be used to obtain seasonalized forecast values, Ysz (or to deseasonalize actual data). Ysz = (SI) Yc

EXPONENTIAL SMOOTHING

Exponential smoothening is a moving-average forecasting technique that weights past data in an Exponential manner so that most recent data carry more weight in the moving average.

With simple Exponential smoothening, the forecast Ft is made up of the last period forecast F_{t-1} plus a portion, α, of the difference between the last periods actual demand A_{t-1} and last period forecast F_{t-1}.1

$$Ft = F_{t-1} + (A_{t-1} - F_{t-1}).$$

Adjusted Exponential Smoothing

Adjusted exponential smoothing models have all the features of simple exponential smoothing models, plus they project into the future (for example, to time period t + 1) by adding a trend correction increment, Tt, to the current period smoothed average, ˆ Ft.

$$\hat{F}_{t+1} = \hat{F}_t + T_1$$

Figure 5.1 depicts the components of a trend-adjusted forecast that utilizes a second smoothing coefficient β . The β value determines the extent to which the trend adjustment relies on the latest difference in forecast amounts (ˆF-ˆFt– 1) versus the previous trend T_{t-1}

Thus:

$$\hat{F}_t = \alpha A_{t-1} + (1-\alpha)\left(\hat{F}_{t-1} + T_{t-1}\right)$$

$$T_t = \beta(\hat{F}_t - \hat{F}_{t-1}) + (1-\beta)T_{t-1}$$

A low β gives more smoothing of the trend and may be useful if the trend is not well-established.

A high β will emphasize the latest trend and be more responsive to recent changes in trend. The initial trend adjustment Tt–1 is sometimes assumed to be zero.

REGRESSION AND CORRELATION METHODS

Regression and correlation techniques quantify the statistical association between two or more variables.

(a) Simple regression expresses the relationship between a dependent variable Y and a independent variable X in terms of the slope and intercept of the line of best fit relating the two variables.

(b) Simple correlation expresses the degree or closeness of the relationship between two variables in terms of a correlation coefficient that provides an indirect measure of the variability of points from the line of best fit. Neither regression nor correlation gives proof of a cause-effect relationship.

5.6.1 Regression

The simple linear regression model takes the form Yc = a + bX, where Yc is the dependent variable and X the independent variable. Values for the slope b and intercept α are obtained by using the normal equations written in the convenient form:

$$b = \frac{\sum XY - n\bar{X}\bar{Y}}{\sum X^2 - nX^2}$$

$$a = \bar{Y} - b\bar{X}$$

STANDARD DEVIATION OF REGRESSION

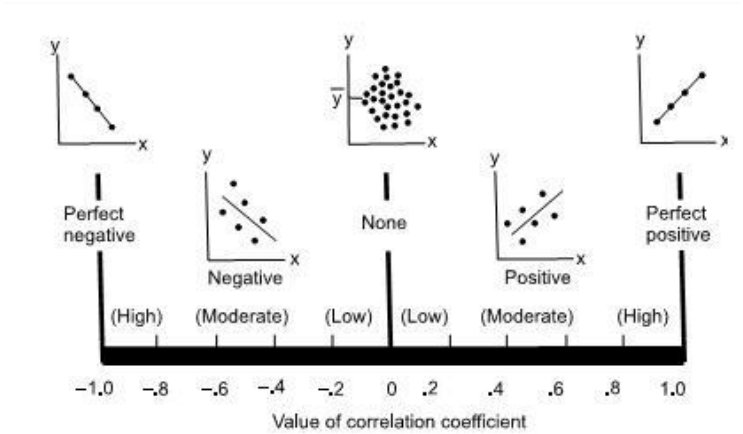
A regression line describes the relationship between a given value of the independent variable X and μy – x the mean of the corresponding probability distribution of the dependent variable Y. We assume the distribution of Y values is normal for any given X value. The point

estimate, or forecast, is the mean of that distribution for any given value of X. The standard deviation of regression Sy – x is a measure of the dispersion of data points around the regression line. For simple regression, the computation of Sy – x has n – 2 degrees of freedom.

$$S_{y-x} = \sqrt{\frac{\sum Y^2 - a\sum Y - b\sum XY}{n-2}}.$$

CORRELATION

The simple linear correlation coefficient r is a number between –1 and + 1 that tells how well a linear equation describes the relationship between two variables. As illustrated in Fig. 5.2 r is designated as positive if Y increases as X increases, and negative if Y decreases as X increases. An r of zero indicates an absence of any relationship between the two variables.



The deviation of all points (Y) from the regression line (Y*c*) consists of deviation accounted for by the regression line (explained) and random deviation (unexplained). Fig. 5.3 illustrates this for one point, Y.

Squaring the deviation we have variation.

Total variation = explained + unexplained

$$\sum(Y-\bar{Y})^2 = \sum(\bar{Y}c - Y)^2 + \sum(Y-Yc)^2$$

The coefficient of determination r^2 is the ratio of explained variation to total variation:

$$r^2 = \frac{\sum(Yc - Y)^2}{\sum(Y-\bar{Y})^2}$$

The coefficient of correlation r is the square root of the coefficient of determination:

$$r = \sqrt{\frac{\sum (Y_C - \bar{Y})^2}{\sum (Y - \bar{Y})^2}}$$

When the sample size is sufficiently large (e.g., greater than 50), the value of r can be computed more directly form:

$$r = \frac{n\sum XY - \sum X.\sum Y}{\sqrt{\left(n\sum X^2 - (\sum X)^2\right)\left[n\sum Y^2 - (\sum Y)^2\right]}}$$

CAPACITY PLANNING

INTRODUCTION

Before products can flow into a market, someone must design and invest in the facilities and organisation to produce them. This chapter concerns the planning of the systems needed to produce goods and services. Capacity Planning for manufacturing and service systems are different. Both must be designed with capacity limitations in mind. The approaches for long-term and short-term capacity planning will help the managers to make best use of resources.

MANUFACTURING AND SERVICE SYSTEMS

Manufacturing and service systems are arrangements of facilities, equipment, and people to produce goods and services under controlled conditions.

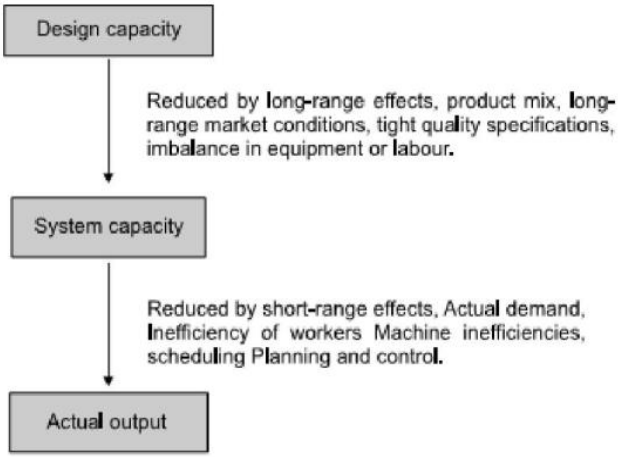
Manufacturing systems produce standardized products in large volumes. This plant and machinery have a finite capacity and contribute fixed costs that must be borne by the products produced. Variable costs are added as labour is employed to combine or process the raw materials and other components.

Value addition will takes place during the production process for the product. The cost of output relative to the cost of input can be measured, as the actual cost is known i.e. productivity is measurable quantity.

Service systems present more uncertainty with respect to both capacity and costs. Services are produced and consumed in the presence of the customer and there is little or no opportunity to store value, as in a finished goods inventory. As a result capacity of service systems like hospitals, restaurants and many other services must be sufficiently flexible to accommodate a highly variable demand. In addition, many services such as legal and medical involves professional or intellectual services judgments that are not easily standardized. This makes more difficult to accumulate costs and measure the productivity of the services.

DESIGN AND SYSTEMS CAPACITY

Production systems design involves planning for the inputs, transformation activities, and outputs of a production operation. Design plays a major role because they entail significant investment of funds and establish cost and productivity patterns that continue in future. The capacity of the manufacturing unit can be expressed in number of units of output per period. In some situations measuring capacity is more complicated when they manufacture multiple products. In such situations, the capacity is expressed as man-hours or machine hours. The relationship between capacity and output is shown in the Figure 3.1.



DESIGN CAPACITY

Designed capacity of a facility is the planned or engineered rate of output of goods or services under normal or full scale operating conditions. For example, the designed capacity of the cement plant is 100

TPD (Tonnes per day). Capacity of the sugar factory is 150 tonnes of sugarcane crushing per day. The uncertainty of future demand is one of the most perplexing problems faced by new facility planners.

Organisation does not plan for enough regular capacity to satisfy all their immediate demands. Design for a minimum demand would result in high utilisation of facilities but results in inferior service and dissatisfaction of customers because of inadequate capacity. The design capacity should reflect management‘s strategy for meeting the demand. The best approach is to plan for some in-between level of capacity.

System/effective capacity: System capacity is the maximum output of the specific product or product mix the system of workers and machines is capable of producing as an integrated whole. System capacity is less than design capacity or at the most equal it because of the

limitation of product mix, quality specification, and breakdowns. The actual is even less because of many factors affecting the output such as actual demand, downtime due to machine/equipment failure, unauthorized absenteeism.

The system capacity is less than design capacity because of long-range uncontrollable factors. The actual output is still reduced because of short-term effects such as breakdown of equipment, inefficiency of labour. The system efficiency is expressed as ratio of actual measured output to the system capacity.

These different measures of capacity are useful in defining two measures of system effectiveness: efficiency and utilization. Efficiency is the ratio of actual output to effective capacity. Utilization is the ratio of actual output to design capacity.

$$\text{Efficiency} = \frac{\text{Actual output}}{\text{Effective capacity}}$$

$$\text{Utilization} = \frac{\text{Actual output}}{\text{Design capacity}}$$

It is common for managers to focus exclusively on efficiency, but in many instances, this emphasis can be misleading. This happens when effective capacity is low compared with design capacity. In those cases, high efficiency would seem to indicate effective use of resources when it does not.

CAPACITY PLANNING

Design of the production system involves planning for the inputs, conversion process and outputs of production operation. The effective management of capacity is the most important responsibility of production management. The objective of capacity management (i.e. planning and control of capacity) is to match the level of operations to the level of demand. Capacity planning is to be carried out keeping in mind future growth and expansion plans, market trends, sales forecasting, etc. It is a simple task to plan the capacity in case of stable demand. But in practice the demand will be seldom stable. The fluctuation of demand creates problems regarding the procurement of resources to meet the customer demand. Capacity decisions are strategic in nature.

Capacity is the rate of productive capability of a facility. Capacity is usually expressed as volume of output per period of time.

Production managers are more concerned about the capacity for the following reasons:

- Sufficient capacity is required to meet the customers demand in time.
- Capacity affects the cost efficiency of operations.

- Capacity affects the scheduling system.
- Capacity creation requires an investment.

Capacity planning is the first step when an organisation decides to produce more or new products.

PROCESS OF CAPACITY PLANNING

Capacity planning is concerned with defining the long-term and the short-term capacity needs of an organisation and determining how those needs will be satisfied. Capacity planning decisions are taken based upon the consumer demand and this is merged with the human, material and financial resources of the organisation.

Capacity requirements can be evaluated from two perspectives—long-term capacity strategies and short term capacity strategies.

1. Long-term capacity strategies: Long-term capacity requirements are more difficult to determine because the future demand and technology are uncertain. Forecasting for five or ten years into the future is more risky and difficult. Even sometimes company‘s today‘s products may not be existing in the future. Long-range capacity requirements are dependent on marketing plans, product development and life-cycle of the product. Long-term capacity planning is concerned with accommodating major changes that affect overall level of the output in long term.

Marketing environmental assessment and implementing the long-term capacity plans in a systematic manner are the major responsibilities of management. Following parameters will affect long-range capacity decisions.

- **Multiple products:** Company‘s produce more than one product using the same facilities in order to increase the profit. The manufacturing of multiple products will reduce the risk of failure.

Having more than on product helps the capacity planners to do a better job. Because products are in different stages of their life cycles, it is easy to schedule them to get maximum capacity utilisation.

- **Phasing in capacity:** In high technology industries, and in industries where technology developments are very fast, the rate of obsolescence is high. The products should be brought into the market quickly. The time to construct the facilities will be long and there is no much time, as the products should be introduced into the market quickly. Here the solution is phase in capacity on modular basis. Some commitment is made for building funds and men towards

facilities over a period of 3-5 years. This is an effective way of capitalizing on technological breakthrough.

• **Phasing out capacity**: The outdated manufacturing facilities cause excessive plant closures and down time. The impact of closures is not limited to only fixed costs of plant and machinery.

Thus, the phasing out here is done with humanistic way without affecting the community. The phasing out options makes alternative arrangements for men like shifting them to other jobs or to other locations, compensating the employees, etc.

2. Short-term capacity strategies: Managers often use forecasts of product demand to estimate the short-term workload the facility must handle. Managers looking ahead up to 12 months, anticipate output requirements for different products, and services. Managers then compare requirements with existing capacity and then take decisions as to when the capacity adjustments are needed.

For short-term periods of up to one year, fundamental capacity is fixed. Major facilities will not be changed. Many short-term adjustments for increasing or decreasing capacity are possible. The adjustments to be required depend upon the conversion process like whether it is capital intensive or labour intensive or whether product can be stored as inventory. Capital-intensive processes depend on physical facilities, plant and equipment. Short-term capacity can be modified by operating these facilities more or less intensively than normal. In labour intensive processes short-term capacity can be changed by laying off or hiring people or by giving overtime to workers. The strategies for changing capacity also depend upon how long the product can be stored as inventory.

The short-term capacity strategies are:

1. **Inventories:** Stock finished goods during slack periods to meet the demand during peak period.

2. **Backlog:** During peak periods, the willing customers are requested to wait and their orders are fulfilled after a peak demand period.

3. **Employment level (hiring or firing):** Hire additional employees during peak demand period and layoff employees as demand decreases.

4. **Employee training:** Develop multi skilled employees through training so that they can be rotated among different jobs. The multi skilling helps as an alternative to hiring employees.

5. **Subcontracting:** During peak periods, hire the capacity of other firms temporarily to make the component parts or products.

6. **Process design:** Change job contents by redesigning the job.

IMPORTANCE OF CAPACITY DECISIONS

1. Capacity decisions have a real impact on the ability of the organisation to meet future demands for products and services; capacity essentially limits the rate of output possible. Having capacity to satisfy demand can allow a company to take advantage of tremendous opportunities.
2. Capacity decisions affect operating costs. Ideally, capacity and demand requirements will be matched, which will tend to minimize operating costs. In practice, this is not always achieved because actual demand either differs from expected demand or tends to vary (e.g., cyclically). In such cases, a decision might be made to attempt to balance the costs of over and under capacity.
3. Capacity is usually a major determinant of initial cost. Typically, the greater the capacity of a productive unit, the greater its cost. This does not necessarily imply a one for-one relationship; larger units tend to cost proportionately less than smaller units.
4. Capacity decisions often involve long-term commitment of resources and the fact that, once they are implemented, it may be difficult or impossible to modify those decisions without incurring major costs.
5. Capacity decisions can affect competitiveness. If a firm has excess capacity, or can quickly add capacity, that fact may serve as a barrier to entry by other firms. Then too, capacity can affect delivery speed, which can be a competitive advantage.
6. Capacity affects the ease of management; having appropriate capacity makes management easier than when capacity is mismatched.

AGGREGATE PRODUCTION (OUTPUT) PLANNING

This plan is the production portion of the business plan and addresses the demand side of the firm's activities by showing the outputs it will produce, expressed in numbers of units of its product groups or families. Since various product groups may be produced at diverse plants, facilities, or divisions, each of them needs its own production plan. The division's aggregate output plan covers the coming 6 to 18 months on a weekly or monthly basis. Planning at this level ignores such details as how many of each individual product, style, color option, or model to produce. The plan recognizes the division's existing fixed capacity and the - company's overall policies for maintaining inventories and backlogs, employment stability, and subcontracting.

Aggregate Capacity Planning*:* A statement of desired output is useful only if it is feasible. This is the role of aggregate capacity planning-to keep capacity Utilization at desired levels and to test the feasibility of planned output against existing capacity. Thus it addresses the supply side of the firm's ability to meet the demand. As for aggregate output plans, each plant, facility, or division requires its own aggregate capacity plan. Capacity and output must be in balance, as indicated by the arrow between them in Figure. A capacity plan translates an output plan into input terms, approximating how much of the division's capacity will be consumed. A product group, for example, usually consumes predictable amounts of capacity such as labor hours of assembly or machine hours for fabrication. Although these basic capacities are fixed, management can manipulate the short-term capacities by the ways they deploy their work force, by subcontracting, or by using multiple work shifts to adjust the timing of overall outputs. As a result, the aggregate planning process balances output levels, capacity constraints, and temporary capacity adjustments to meet demand and utilize capacity at desired levels during the coming months. The resulting plan sets limits on the master production schedule.

AGGREGATE PLANNING PROCESS

The process consists of four basic considerations as follows:

1. Concept of Aggregation starts with a meaningful measure of output. In a single product output organization there is no problem with the output measure. Many organizations have multiple products and it is difficult to find a common factor of measure of output.

For e.g. steel producer can plan in terms of tons of steel, gallons of paint in case of paint industry.

Service organizations such as transport system may use passenger miles as a common measure, health care facilities may use patient visits, and educational institutes may use student to faculty contact ratio in terms of hours as a reasonable measure.

You may recall that a group of products or services that have similar demand requirements and common processing, labour and materials requirements is called a Product Family. Therefore a firm can aggregate its products or services into a set of relatively broad families, avoiding too much detail at the planning stage.

For example consider the Bicycle manufacture that has aggregated all products into two families: mountain bikes and road bikes. This approach aids production planning for the assembly lines in the plants

2. Goals for aggregate planning

There are number of goals to be satisfied

• It has to provide the overall levels of output, inventory and backlogs dictated by the business plan

• Proper utilization of the plant capacity. It should not be underutilized because it is waste of resources.

It is better to operate at a near full capacity.

• The aggregate plan should be consistent with the company‘s goals and policies regarding its employees.. A firm may like to have employee stability or hire and layoff strategy. Other firms change employees freely as the output level is varied throughout the aggregate planning horizon.

3. Aggregate Demand Forecasts

The benefits of aggregate planning depend on the accurate forecasting. The forecasting models presented in Chapter 3 can be used to forecast demand for product groups as well as individual products.

4. Interrelationships among decisions Here the managers must consider the future consequences of current decisions. This is important mainly due to the fact that output plans are developed for a long period of time.

Master Production Scheduling (MPS)

The purpose of master production scheduling is to meet the demand for individual products in the product group. This more detailed level of planning disaggregates the product groups into individual products and indicates when they will be produced. The MPS is an important link between marketing and production. It shows when incoming sales orders can be scheduled into production, and when each shipment can be scheduled for delivery.

It also takes into account current backlogs so that production and delivery schedules are realistic.

Rough-Cut Capacity Planning: Rough-cut capacity planning (sometimes called resource requirements planning) is done in conjunction with the tentative master production schedule to test its feasibility in terms of capacity before the MPS is finally settled. This step ensures that a proposed MPS does not inadvertently overload any key department, work center, or machine, making the MPS unworkable. Although the check can apply to all work centers, it is typically applied only to the critical ones that are most likely to be bottlenecks. It is a quick

and inexpensive way to find and correct gross discrepancies between the capacity requirements (in direct labor hours, for example) of the MPS and available capacity.

Detailed Capacity Planning: Detailed capacity planning, also called capacity requirements planning, is a companion process used with MRP to identify in detail the capacity required to execute the MRP. At this level, more accurate comparisons of available and needed capacity for scheduled work loads are possible.

Shop Floor Control: Shop .floor control coordinates the weekly and daily activities that get jobs done. Individual jobs are assigned to machines and work centers (loading), the sequence of processing the jobs for priority control is determined, start times and job assignments for each stage of processing are decided (detailed scheduling), and materials and work flows from station to station are monitored and adjusted (expediting). All these activities are co-coordinated into smooth flow, especially when there are delays and priorities arise, last minute adjustments of outputs and capacities is known as short term capacity control.

Capacity Requirements Planning (CRP) is the process of determining what personnel and equipment capacities (times) are needed to meet the production objectives embodied in the master schedule and the material requirements plan. MRP focuses upon the priorities of materials, whereas CRP focuses primarily upon time. Although both MRP and CRP can be done manually and in isolation, they are typically integrated within a computerized system, and CRP (as well as production activity control) functions are often assumed to be included within the concept of an MRP system. Computerized MRP systems can effectively manage the flow of thousands of components throughout a manufacturing facility.

CRP INPUTS AND OUTPUTS

Capacity is a measure of the productive capability of a facility per unit of time. In terms of the relevant time horizon, capacity management decisions are concerned with the following:

l. Long range-resource planning of capital facilities, equipment, and human resources.

2. Medium range-requirements planning of labor and equipment to meet MPS needs.

3. Short range-control of the flow (input-output) and sequencing of operations.

Capacity-requirements planning (CRP) applies primarily to medium-range activities. The CRP system receives planned and released orders from the material-requirements planning system and attempts to develop loads for the firm‘s work centers that are in good balance

with the work-center capacities. Like MRP, CRP is an iterative process that involves planning, revision of capacity (or revision of the master schedule), and re planning until a reasonably good load profile is developed. Planned-order releases (in the MRP system) are converted to standard hours of load on key work centers in the CRP system.

Following are the essential inputs and outputs in a CRP system:

Inputs	Outputs
• Planned and released orders from the MRP system • Loading capacities from the work-center status file • Routing data from the routing file •Changes that modify capacity, give alternative routings, or alter planned orders	• Verification reports to the MRP system •Load reports of planned and released orders on key work centers • Rescheduling data to the MPS • Capacity modification data

Material Requirements Planning (MRP) is a computer-based technique for determining the quantity and timing for the acquisition of dependent demand items needed to satisfy the master schedule requirements. By identifying precisely what, how many, and when components are needed, MRP systems are able to reduce inventory costs improve scheduling effectiveness, and respond quickly to market changes.

Following are some of the terminology used to describe the functioning of MRP systems.

• **MRP:** A technique for determining the quantity and timing dependent demand items.

• **Dependent demand:** Demand for a component that is derived from the demand for other items.

• **Parent and component items:** A parent is an assembly made up of basic parts, or components. The parent of one subgroup may be a component of a higher-level parent.

• **Bill of materials:** A listing of all components (subassemblies and materials) that go into an assembled item. It frequently includes the part numbers and quantity required per assembly.

• **Level code:** The level on which an item occurs in the structure, or bill-of-materials format.

• **Requirements explosion:** The breaking down (exploding) of parent items into component parts that can be individually planned and scheduled.

• **Time phasing:** Scheduling to produce or receive an appropriate amount (lot) of material so that it will be available in the time periods when needed-not before or after.

• **Time bucket:** The time period used for planning purposes in MRP-usually a week.

• **Lot size:** The quantity of items required for an order. The order may be either purchased from a vendor or produced in-house. Lot sizing is the process of specifying the order size.

• **Lead-time offset:** The supply time, or number of time buckets between releasing an order and receiving the materials.

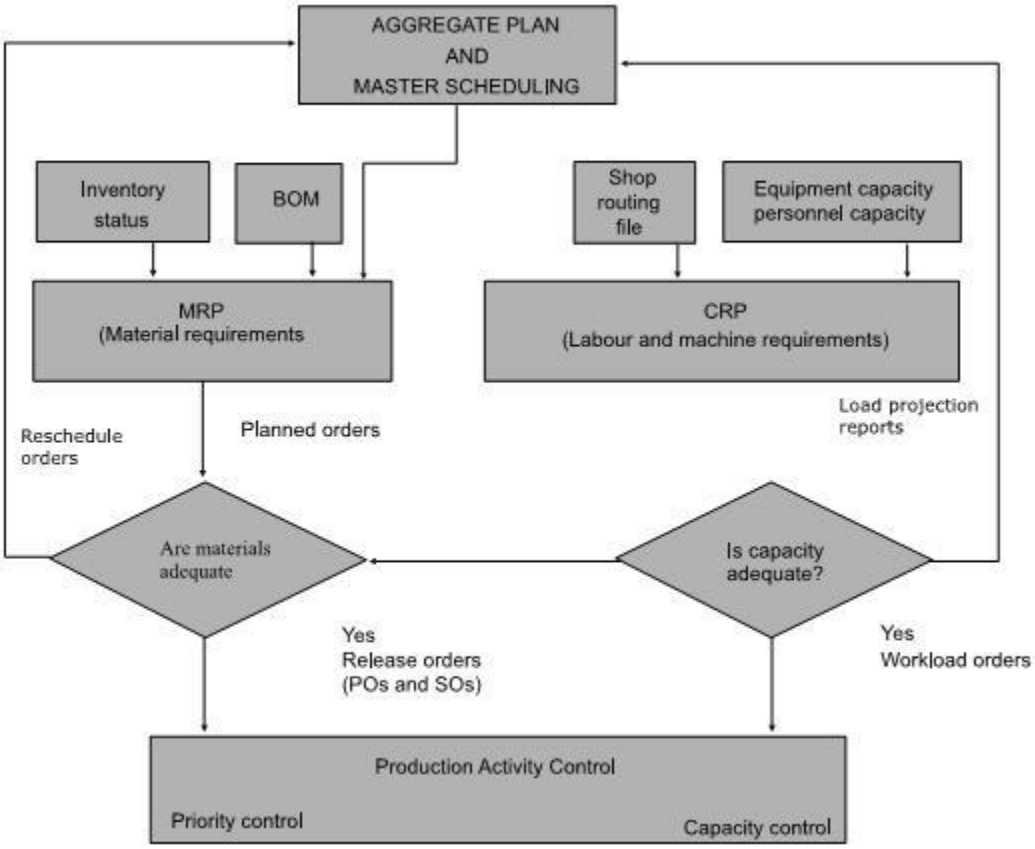


Figure 9.1 describes MRP and CRP activities in schematic form. Forecasts and orders are combined in the production plan, which is formalized in the master production schedule (MPS). The MPS, along with a bill-of-material (BOM) file and inventory status information, is used to formulate the material requirements plan. The MRP determines what components are needed and when they should be ordered from an outside vendor or produced in-house. The CRP function translates the MRP decisions into hours of capacity (time) needed. If materials, equipment, and personnel are adequate, orders are released and the workload is assigned to the various work centers. End items, such as TV sets, have an independent demand that is closely linked to the ongoing needs of consumers. It is random but relatively constant.

Dependent demand is linked more closely to the production process itself. Many firms use the same facilities to produce different end items because it is economical to produce large lots once the set-up cost is incurred. The components that go into a TV set, such as 24-inch picture tubes, have a dependent demand that is governed by the lot size. Dependent demand is predictable.

MRP systems compute material requirements and specify when orders should be released so that materials arrive exactly when needed. The process of scheduling the receipt of inventory as needed over time is time phasing.

MRP is a system of planning and scheduling the time-phased materials requirements for production operations. As such, it is geared toward meeting the end-item outputs prescribed in the master production schedule.

MRP OBJCTIVES AND METHODS

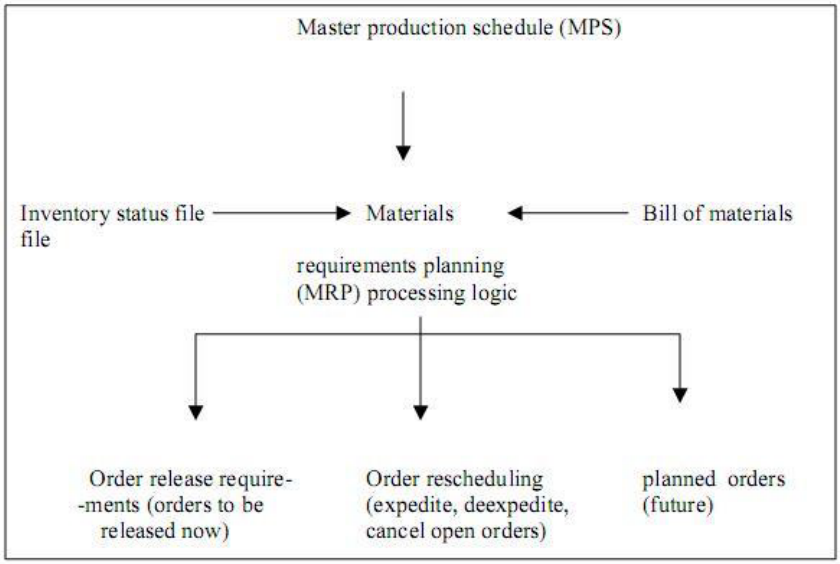
MRP provides the following:

1. Inventory reduction: MRP determines how many of component is needed and when, in order to meet the master schedule.

2. Reduction in production and delivery lead times: MRP identifies materials and components quantities, timings, availabilities, and procurement and production actions required to meet delivery dealings.

3. Realistic commitments: Realistic delivery promises can enhance customer satisfaction. By using MRP, production can give marketing timely information about likely delivery times to prospective customers.

4. Increased efficiency: MRP provides close coordination among various work centers as products progress through them.



MRP INPUTS AND OUTPUTS

The essential inputs and outputs in an MRP system are listed below:

Inputs	Outputs
• MPS of end items required. • Inventory status file of on-hand and on-order items, lot sizes, lead times, etc. • Product structure (BOM) file of what components and subassemblies go	• Order release data to CRP for load profiles • Orders to purchasing and in-house production shops. • Rescheduling data to MPS. • Management reports and inventory updates.

A bill of materials (BOM) is a listing of all the materials, components, and subassemblies needed to assemble one unit of an end item. Major function of the bill of materials is to provide the product structure hierarchy that guides the explosion process. Different methods of describing a BOM are in use. Figure 9.2 shows (a) a product structure tree, and (b) an indented BOM. Both are common ways of depicting the parent-component relationships on a hierarchical basis. Knowledge of this dependency structure reveals clearly and immediately what components are needed for each higher level assembly. A third method (c) is to use single-level bills of material.

MANUFACTURING RESOURCES PLANNING (MRPII)

Historically, MRP systems typically were developed on a segregated basis, rather than as part of highly integrated information system. More recently, however, companies are beginning to logically relate many of their information subsystems to the MRP system. Bills of materials data, for example, can be shared with an engineering information system data base; order release and order receipts data can be shared by the order billing and accounts payable information systems; and inventory status data from MRP can be part of marketing or purchasing information systems. This type of information integration, in fact, is exactly the impetus for a new generation of manufacturing planning and control systems.

Manufacturing resource planning (MRP II, or ―closed loop‖ MRP) is an integrated information system that steps beyond first-generation MRP to synchronize all aspects (not just manufacturing) of the business. The MRPII system coordinates sa les, purchasing, manufacturing, finance, and engineering by adopting a focal production plan and by using one unified data base to plan and update the activities in all the systems.

As shown in figure the process involves developing a production plan from the business plan to specify monthly levels of production for each product line over the next one to five years. Since the production plan affects all the functional departments, it is developed by the consensus of executives and becomes their game plan for operations. The production

department then is expected to produce at the committed levels, the sales department to sell at these levels, and the finance department to ensure adequate financial resources for these levels. Guided by the production plan, the master production schedule specifies the weekly quantities of specific products to be built. At this point a check is made to determine whether the capacity available is roughly adequate to sustain the proposed master schedule. If not, either the capacity or the master schedule must be changed. Once settled, the master schedule is used in the MRP logic, as previously described, to create material requirements and priority schedules for production. Then, an analysis of detailed capacity requirements determines whether capacity is sufficient for producing the specific components at each work center during the scheduled time periods.

If not, the master schedule is revised to reflect the limited available capacity. After a realistic, capacity feasible schedule is developed, the emphasis shifts to execution of plan: purchase schedules and shop schedules are generated. From these schedules, work center loadings, shop floor control, and vendor follow-up activities can be determined to ensure that the master schedule is met.

One use of the MRPII system is to evaluate various business proposals. If, for example, the output of product X increases by 20 percent in weeks 15 to 20 and that of Y decreases by 15 percent in weeks 10 to 15, how would operations and profitability be affected? the system can simulate how purchases and, hence, accounts payable are affected? The system can simulate how purchases and, hence, accounts payable are affected, when delivers to customers and accounts receivable occur, what capacity revisions are needed, and so on. The company-wide implications of the proposed change can be evaluated, and various departments can be coordinated according to a common purpose.

UNIT 3

DESIGN OF PRODUCT, SERVICE AND WORK SYSTEMS

Introduction:

Management has many options in the selection, definition, and design of goods and services. Product selection is choosing the good or service to provide customers or clients. For instance, hospitals specialize in various types of patients and various types of medical procedures. They select their product when they decide what kind of hospital to be. A hospital‘s management may decide to operate a general purpose hospital, a maternity hospital, or even a hospital specializing in heart diseases. Numerous other options exist for hospitals, just as they exist for restaurants and automobile companies.

Product decisions

Product design is the mother of all operations processes in an organisation. The processes for manufacture, the planning of production, the processes and checks for quality depend upon the nature of the product. One may say that it all starts with the design of the product. Even the logistics or plain shipment of the product depends upon how or what the product has been designed for. Design gives the blueprint. When the design engineer keys in the computer aided design or when a product design artist draws lines on a sheet of paper, it starts a train of activities.

Product decisions are fundamental and have major implications throughout the operations function.

They influence capital equipment cost, layout design, space requirements, the skills of people hired and trained, materials, and the processes used.

Factors Influence In Generation Of New Product Opportunities

Product selection, definition, and design take place on a continuing basis because so many new product opportunities exist.

Five factors influencing market opportunities are

1. Economic change, which brings increasing levels of affluence in the long run but economic cycles and price changes in the short run. For instance, in the long run, more and more people can afford an automobile, but in the short run, a recession may alter the demand for automobiles.

2. Sociological and demographic change, which may appear in such factors as decreasing family size. This alters the size preference for homes, apartments, and automobiles.

3. Technological change, which makes possible everything from home computers to mobile phones to artificial hearts.

4. Political/legal change, which brings about new trade agreements, tariffs, and government contract requirements.

5. Other changes, which may be brought about through market practice, professional standards, suppliers, and distributors.

PRODUCT ANALYSIS Many factors have to be analyzed in connection with development and design, factors varying in character and complexity, factors affiliated with different fields in production and industrial engineering. Some of these may be grouped as follows:

1. Marketing aspect

2. Product characteristics

(i) Functional aspect,

(ii) Operational aspect,

(iii) Durability and dependability aspects, and

(iv) Aesthetic aspect.

3. Economic analysis

(i) The profit consideration,

(ii) The effect of stardardization, simplification, and specialization, and

(iii) The break-even analysis.

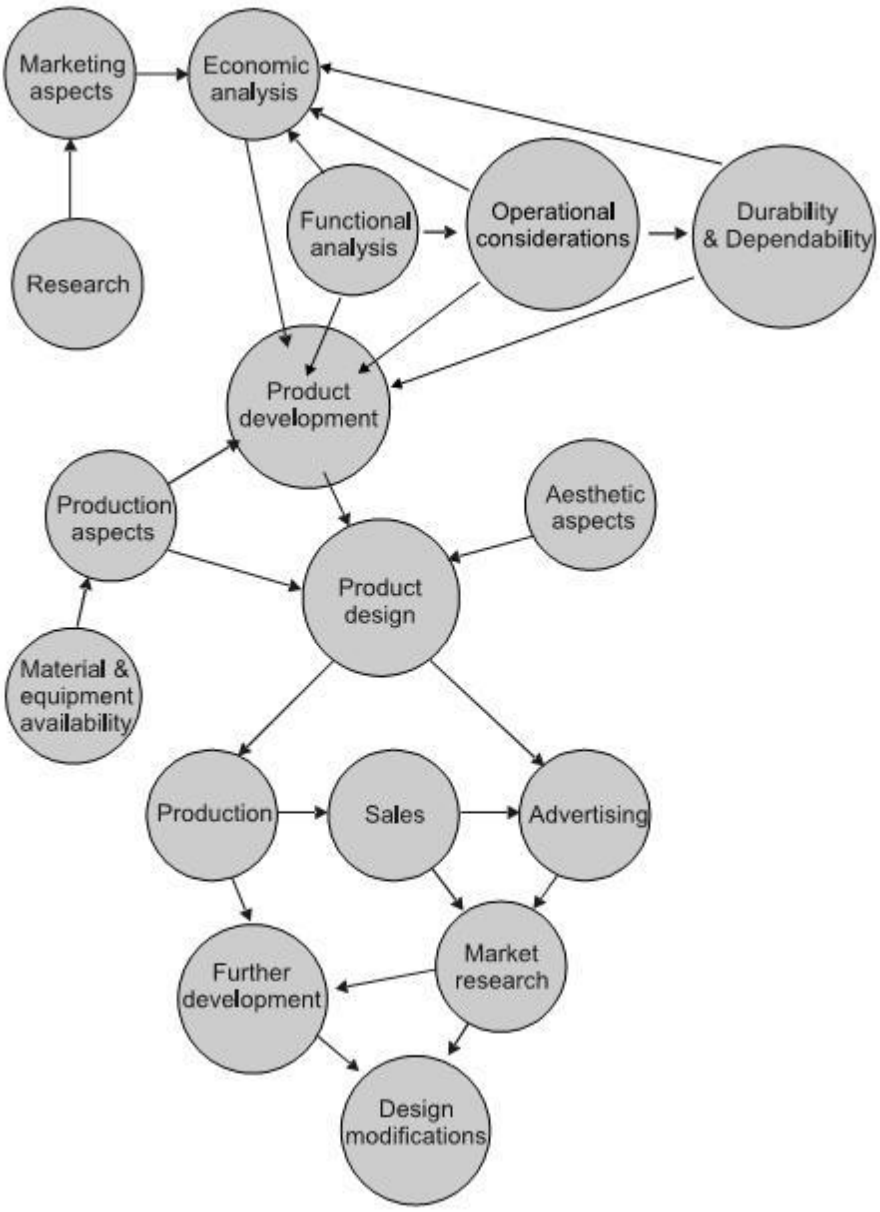
4. Production aspect

All these factors are interrelated and each presents many issues that have to be carefully considered, as indicated by Figure 6.1. Market research may guide product engineers in their work to improve existing products or to develop new ones. The design and its characteristics have to undergo an economic analysis and must be studied in the light of available production facilities and techniques. A costing analysis is naturally dependent on the sales volume; hence the suggested design has to be re-evaluated by market research so that a sales forecast can be worked out. This expected sales volume provides the basis for a further study from the production methods aspect, and the economic analysis has to be rechecked and perhaps modified. Thus product development and design is an excellent example of interdependence of a multitude of factors that have to be reconciled and integrated into a final composition.

1. Marketing Aspect

First, it is necessary to establish that the proposed product will satisfy a demand in the market, that what it is supposed to do and the services it can offer are both desirable and acceptable. If no consumption is envisaged, there is no point in proceeding with product design. The demand for the product in the market may already exist, and its volume can then

be assessed by consumer research and sales figures for identical or similar commodities. Demand can also be created with the introduction of a new product, either by filling in a gap in the market or by offering new properties, such as novelty, appearance, or some other specific merits.



The volume of such a demand is more difficult to forecast. Market research is a useful tool in these cases, but experience and sound judgment are required to evaluate and apply the results of such research, and in some cases a certain amount of speculation is inevitable.

The volume of demand is a function of a multitude of factors, some of which are closely related to local conditions and are sometimes difficult to define or measure. It is therefore essential for an enterprise to keep in touch with the market and ―feel‖ its trends, especially

when this market is remote and different in character from the local one. This is of particular importance to firms depending on export markets for the distribution of their products.

If we analyze, for example, the case of an American manufacturer of automobiles, we shall find that the percentage of output destined for export is rather small, and design policy would therefore be mainly dictated by American tastes and preferences. A British manufacturer, however, who sells a substantial proportion of automobiles outside Great Britain, has to watch carefully the trends in export markets in order to try and amalgamate the requirements and tastes of the various foreign and home markets in an acceptable design.

Another pertinent question related to product design is : Should the customer get what he wants or should he be offered what he is supposed to want? Basically this is an economic question. If management wants to achieve maximum satisfaction and sets itself as a target to supply the customer with what he wants, it may be faced with the possibility of having to produce an infinite variety of models to suit every taste.

On the other hand, were management to ignore the customer‘s wishes or to maintain that he does not really know what he wants and should therefore be told what is good for him, the designer‘s task would become far simpler, but the sales department would have to face an unpredictable market.

In practice, product design is a result of some sort of compromise between infinite variety on one hand and the designer‘s concept of the ideal design on the other. In order to try selling this compromise to potential customers, management resorts to an advertising campaign the policy of which is dependent on the characteristics of the ―compromised design‖ and on how far it conforms to, or differs from, the expressed desires of the market to which such a campaign is directed. Generally, the main objective of advertising is to expand the market, this being achieved by:

- Providing general information about the existence of the product.
- Providing technical information about its functional characteristics or utilitarian purposes.
- Drawing the customer‘s attention to those attributes of the product which he wants.
- Winning undecided customers by exhibiting possible attractions (such as colour, design, novelty, and price) that may persuade him to prefer the product to one offered by competitors.
- Creating demand among a passive population of customers.
- Educating the customer, or telling him what he should want.

Apart from these direct techniques, management may have some additional aims, such as increasing the prestige of the firm as a whole, banking on the popularity of one product to strengthen or introduce another or to publicize one aspect of the firm‘s activity for the purpose of raising money or deviating attention from other activities, and so on. Once the design features of a product have been ascertained, appropriate advertising methods can be selected.

THE PRODUCT CHARACTERISTICS

Functional Aspect

When the marketing possibilities have been explored, the functional scope of the product has to be carefully analyzed and properly defined. The definition of the objective itself rarely tells us very much about the functional scope envisaged. A washing machine, for example, has a clearly defined objective: to wash clothing. This does not state, however, how the washing should be carried out, whether the machine should be capable of heating the water prior to washing, whether rinsing or drying, or both, are to be done by the machine, and if so by what method, and what should the proportion be between automatic functioning and manual supervision. A functional analysis of this kind obviously affects the design of the machine, its complexity, its appearance, and its price. Sometimes functional aspects are detachable, and usage can be left to the customer‘s decision. A steam iron is a case in point. The additional function of dampening the cloth when required, prior to or during ironing, is incorporated in the steam iron, the main duty of which is to iron the cloth. The customer can decide whether and when to exploit this characteristic of the apparatus. There is a trend to offer functional versatility of the product, thereby increasing the range of applications and sometimes combining several tools in one. A food mixer, for example, allows for a large number of attachments to be added for a variety of duties.

Basically the mixer housing contains a power unit and a speed regulator, but it has to be designed so as to serve all the attachments, and the customer has to decide and define for himself the functional scope to be compatible with his needs, his taste, and his pocket. Household power-tool sets are designed on very much the same principle: The hand drill is the basic unit, and with attachments it can become a table drill, a lathe, a polisher, a hedge cutter, etc. Versatility of production machinery may quite often result in substantial savings in floor space and capital expenditure, and this may become one of the fundamental factors affecting design policy. Another example of versatility in design is shown in Figure.

Operational Aspect

After determining the functional aspect, the operational aspect has then to be considered. Not only must the product function properly, it must be easy to handle and simple to operate. Sometimes it has to be adaptable to various operational conditions, and very often it is subjected to varying degrees of skill of potential operators. The designer‘s problem becomes all the more critical with the trend for increased versatility because this characteristic implies using basic attachments as elements for building suitable combinations for specific purposes. This requires a certain amount of operator intelligence and skill, which increases with the complexity of the machine. The scarcity of skill is a prohibitive limitation in this respect on the product designer.

The ‘get ready‘ stage before the operation proper and the _put away‘ time (including cleaning) should be carefully analyzed with respect to the excepted skill of the operator. Too often one finds ingenious gadgets (for example, in the field of household equipment) that are capable of performing an operation in a fraction of the time normally required but which involve such complicated preparations or such lengthy cleaning and ‘put away‘ subsequent operations, that the ratio of net machine time to overall machine time becomes absurdly small. The beneficial features attributed to the gadget in such cases are rather questionable.

Versatility of equipment should also be analyzed in this light. Especially when subsequent operations are to be carried out with the aid of different attachments, the designer should always bear in mind the time required for an operator to perform the change over and should make certain that this time is in reasonable proportion to the operation time.

Durability and Dependability

These are two factors closely related to the selection of materials and class of workmanship and hence to the design of the product and the economical analysis of its cost. Quality is not always a simple characteristic to define, but durability and dependability are two factors that often determine quality and have to be carefully considered by the designer. Durability is defined mainly by the length of the active life, or endurance, of the product under given working conditions, but a measure of the product capability to idle or withstand storage is also often considered in assessing durability. Durability need not always be associated with selection of good materials. The actual working life of a match or a rocket motor may be rather limited, but that does not mean that materials for these articles may be of low quality. An additional criterion, therefore, has to be considered, that of dependability, or the capability of the product to function when called upon to do its job. Returning to our matches, dependability may be related to the number of duds in a box, and while the manufacturer is

eager to reduce this number to a minimum, he need not choose the very best raw materials to ensure that not even one match will fail. Dependability of rocket motors, however, may be more rigidly defined, and first class materials are chosen in spite of the short active life that is envisaged for them in some applications.

Another aspect of durability is that of maintenance and repair. The amount of repair and preventative maintenance required for some products is closely related to quality and design policy. This is of particular importance when the equipment is supposed to operate continuously and when any repair involves a loss of running time.

Problems of convenience and accessibility in operating the equipment have already been discussed, and the same remarks are valid for maintenance and repair. Easy accessibility is a fundamental principle in a sound design, and thorough knowledge on the part of the operational durability, dependability, and maintenance requirements of the product are absolutely essential to ensure a well- balanced design within the policy outlined by higher management.

Aesthetic Aspect

In what way does the appearance of a product affect its design? In most cases, where the functional scope, durability, and dependability have already been defined, the aesthetics are mainly concerned with molding the final shape around the basic skeleton. This molding of shape may very often be severely limited in scope, and what finally emerges is sometimes termed a _functional shape‘. The view that functional shape is necessarily divorced from aesthetics, especially where engineering structures or equipment are concerned, is well-exemplified by bridges, locomotives, or machines of the late nineteenth or early twentieth century (see, for example, Fig. 6.1. However, a study of the gradual changes in shape of these objects in the past few decades would convince us that there has been an increasing recognition of the role of aesthetics in design. This is perhaps partly due to man‘s aesthetic taste being reconciled to accepting these objects as an integral part of the landscape or everyday life, thereby leading to a modification of the original attitude that these ―Monstrosities‖ are hopelessly ugly and should be left alone.

Functional shape is a concept in its own right among designers. Those who believe in functional shape argue that compatibility of function with shape is logical and should therefore be accentuated and exploited, rather than covered up. A standard lamp is first and foremost a lamp and not a flying saucer, and there is nothing wrong with its looking like a lamp. This approach is referred to in Fig. 6.2, where the aesthetic aspects are dealt with at the design stage, after all the other aspects of the proposed product have been analyzed.

In some cases, however, molding of shape may have financial implications; for instance, when special materials have to be used or added to those basically required from the functional point of view or when additional processes are involved. Such cases will call for a careful cost analysis of the aesthetic aspects.

In extreme cases, aesthetics are the governing factor in design and completely dominate it. This is especially true for many consumer goods, such as automobiles and household equipment, or fashion goods. The functional scope, though more or less defined and accepted, may also be widened136

Operations Management to accentuate the novelty of the new model. But the idea of the new design starts with the concept of its shape, from which the idea evolves and grows. The technical considerations have to be somehow fitted in at a later stage, this being in complete contrast to the conventional sequence shown in Fig.

When styling is a dominant factor in product design, it is often used as a means to create demand. Changes in fashion and taste, evolution of form, and the introduction of new ideas quickly outdate previous designs. If the market is psychologically receptive and eager to discard former designs in favour of new ones, styling becomes a race against time, a race that determines the salability of the product.

Many tools can be utilized by the designer to bring out aesthetic characteristics. Some of these are:

1. **Use of the special materials**, either for the parts of the housing or as additional decorations. Notable is the use of chromium strips, plastics, wood, glass, and fabrics for the purpose.

2. **Use of colour,** either natural colour of the material concerned or by use of paints, plating, spraying, or even lighting. Composition and contrast of colours is of great importance to the industrial designer in facilitating convenient operation and attractive appearance.

3. **Texture supplements colour,** either by appropriate treatment of the given surfaces or coatings. Surface finish and requirements of brightness as determined by styling may in turn affect the production processes in the finishing stages.

4. **Shape** denoted by outer contours and similarity to familiar objects. Shape can be exploited to accentuate particular features, to create a sense of spaciousness or illusions of size, richness and dependability.

5**. Line** is used to break the form, also for the purpose of emphasizing parts of it, or to give a sense of continuity, graciousness, and stability.

6**. Scaling** the product, either to a blown-up size or to a small size (modeling). This creates novelty and a sense of completeness. The success of styling of some popular small automobiles in Europe may be partly due to the designer‘s talent in creating a feeling of still having the full-size version, with all its features.

7. **Packaging,** especially for small items. Novelty and attractiveness of packaging is often transferred in the mind of the customer, attributing perhaps non-existent values to the contents. In extreme cases packaging may assume an appreciable portion of the total production costs and become the center of the design project. Aesthetic molding, especially when governed by the selection of material, colour, texture, and sometimes even line, has great economic advantages, since great variety can be achieved at a comparatively low cost. The basic product remains the same, and variety is obtained by finishing processes alone. Henry Ford‘s maxim that the customer may choose any colour he likes, provided it is black, is no longer valid. Modern production control techniques allow for a vast number of combinations of colours and textures to be offered with little difficulty. Aesthetics have been fully recognized as an integral part of design, and no designer worth his mettle can afford to ignore their implications, their tools, and their benefits.

ECONOMIC ANALYSIS

As shown in Figure 6.1, an economic analysis is the key to management decision in product design policy. Having obtained sufficient information about customers‘ requirements and market potentialities on the one hand and a detailed study about the functional, operational, and quality aspects of the proposed product on the other, the economic analysis can proceed by seeking an answer to the following questions:

- What capital expenditure is required for manufacturing the new product?
- What total production costs per piece are envisaged?
- What is the reasonable margin of profit that can be expected?
- Do the price (= total costs + profit) and the features of the product render it competitive in the market ?
- In what numbers is the product expected to be sold?

Here, again, the interdependence of variables should be strongly emphasized. Not one single question in this list can be isolated and solved independently of the others. The economic analysis is in fact a cyclic and repetitive procedure. Each question is weighted in the light of the answer and the data provided by the previous question, and all the answers are checked

when their turn comes again to be re-evaluated in the following cycles, until a state of equilibrium is reached and no further modifications to these answers are required.

Profit and Competitiveness

The measure of competitiveness of the product corresponds to the portion of the market it succeeds in capturing. This is largely dependent on the value the customer is prepared to put on the product, and on the ratio of this value to the price. As customer assessment of value is not universally uniform but subject to preference of features, performance, or taste, ratios of values to prices vary with customers. A state of equilibrium is formed in which the market is divided between different preferences. This equilibrium may change: If the ratio of value to price of the product becomes more favourable, when compared with other products, the product increases its portion of the market and becomes more competitive.

Such an equilibrium is shown in Fig. 6.3 where the total costs include set-up, materials, overheads, storage, and distribution. The total profit is determined by the margin of profit per unit and by the sales volume. If the organization seeks to increase its profit, it can try one of the following methods 6.3:

(a) Increase the margin of profit per unit, hence the sales price, but leave the total production to costs unchanged. If such a course would not affect the sales volume, the total profit would be proportional to the increase in the margin of profit per unit. Such an increase, however, can upset the market equilibrium unfavourably, in that both the ratio of customers‘ value of the product to its price will deteriorate and the products of competitors will become more attractive. The market may shrink, and the total profit, far from attaining the expected value, may in extreme cases fall below its original level.

(b) Leave the total costs unchanged, but try to improve the ratio of value to price and thus widen the market. This can be done

(1) by producing a better or more attractive product at the same cost,

(2) by launching an intense advertising campaign in order to boost the customer‘s assessment of the product value, or

(3) by reducing the sales price at the expense of the margin of profit per unit, in the hope that the market will expand enough to increase total profit. Too marginal a profit per unit is, however, undesirable, as it allows little protection from possible fluctuations in the market, and even slight instabilities may turn a small profit into a sizeable loss.

(c) Reduce the total production costs and pass some of the benefit to customers in the form of reduced sales prices.

If both the profit per piece and the size of the market increase, a substantial improvement in total profits will be achieved. This course calls for a continuous search after better methods, better processes, better materials and their utilization, and better management to reduce overheads. There are, however, some limitations to the rate of improvement one can attain, such as basic labour and material costs and limited resources or credit hampering expenditure on new equipment and machines. Minimum requirements of quality should also be studied and met, as a reduction in price at the expense of quality is easy enough; customer‘s assessment of the product value, however, deteriorates accordingly. But reducing production costs and thereby expanding the market, while sustaining accepted quality standards, offers a challenge to the production engineer. Probably the most characteristic feature of this process is that it is both dynamic and continuous, that each success is a further advance along the spiral of increasing productivity and standard of living.

The Three S’s

The three S‘s refer to standardization, simplification and specialization—three related subjects that are at the root of any economic analysis of product design. The three S‘s can be defined as follows:

Standardization is the process of defining and applying the ―conditions‖ necessary to ensure that a given range of requirements can normally be met with a minimum of variety and in a reproducible and economic manner on the basis of the best current technique.

Simplification is the process of reducing the number of types of products within a definite range.

Specialization is the process whereby particular firms concentrate on the manufacture of a limited number of products or types of products.

The three processes are usually linked together and develop as a logical sequence. From a wide range of requirements it is first necessary to sort out the essential features, define them, and then work out in a scientific manner, the minimum variety required to meet these essentials. This is a process of standardization, and it is mainly an engineering process. Within a given range, whether covered by standards or not, a process of simplification can be carried out with the view of reducing the variety of products or materials that are produced or purchased. This is both an economic and an engineering process, and specialization is one of its natural outcomes.

Production Aspect

Last but not least in the list of factors influencing design is the production aspect. The product need not only be well-planned on the drawing board, but its design must also be capable of being eventually translated into palpable fact. The designer must therefore face a multitude of practical production problems. ―Design for production‖ ha s become a motto among designers. The following three aspects of production engineering have to be weighed:

1. Selection of processes that will be the most suitable and economical for the purpose. Such a selection will have to consider:

(i) The production quantities involved. Some processes are very expensive to operate unless used for a suitable production run.

(ii) Utilization of existing equipment. Such considerations may override acquisition of equipment for an ideally more suitable process.

(iii) Selection of jigs and fixtures and other production aids, the use of which may affect the design of components.

(iv) Sequence of operations and methods for sub-assembling and assembling.

(v) Limitation of skill. The selection of a process must be compatible with available skill and sometimes may be solely governed by it. Mechanized and push button equipment is particularly suitable to non-skilled or semiskilled operators, but it is usually expensive to install and must be justified by long runs.

(vi) Application of new production processes. The designer has to consider not only conventional techniques but also the latest developments and research into newer production methods.

2. Utilization of materials and components with the view of

(i) Selection of materials having appropriate specifications.

(ii) Selection of method or design to reduce waste and scrap.

(iii) Using standard components and assemblies.

(iv) Having interchangeability of components and assemblies with in the product.

3. Selection of appropriate workmanship and tolerances that satisfy quality requirements, but which are at the same time compatible with the precision and quality that can be attained through the available processes. Specification of quality may also affect the selection of processes.

A FRAMEWORK FOR PROCESS DESIGN

Process design can be viewed as an interactive exercise. That is, problems are solved one at a time and sequentially; then after each stage, or perhaps after several stages, the previous stages are reexamined to see if later steps have affected the best way in which these steps should have been designed. This procedure is illustrated in Fig. 6.13.

6.4.1 Product Planning

Product planning serves as an input to process design. However, in most cases the responsibility for this phase rests with groups, such as marketing and engineering, which are generally found outside the domain of process design. It is early in this stage that the perceived needs of the consumer are identified.

If the process is service-oriented, these needs will be reflected in the proposed quality, speed, cost, and reliability of the service. If, on the other hand, the process will be manufacturing-oriented, then these needs will be reflected in the product‘s proposed quality, cost, function, reliability and appearance.

When these service or product parameters are transformed into a product, design, it is essential that a cross-functional alliance between product, planning and process design groups be established in order to ensure that the product objectives can indeed be profitably met. Otherwise only local goals may be pursued. For example, the marketing and engineering departments working alone may design a product which is very costly to manufacture and very difficult to service. In short, the transformation process should be considered well before the design is finalized. This can usually be accomplished in the following way:

Information from the product development stage can be directed to those responsible for process R&D.

They in turn can determine if the process capability for this product now exists within the firm, whether it exists outside the firm or whether research and development effort would be necessary to meet the specified objectives. Rough cost estimates would also be made for each alternative identified. In large firms a special process R&D department may be organized just for this purpose. In smaller firms this function is less formalized and more reliance would be placed on outside suppliers of processing equipment and company engineers.

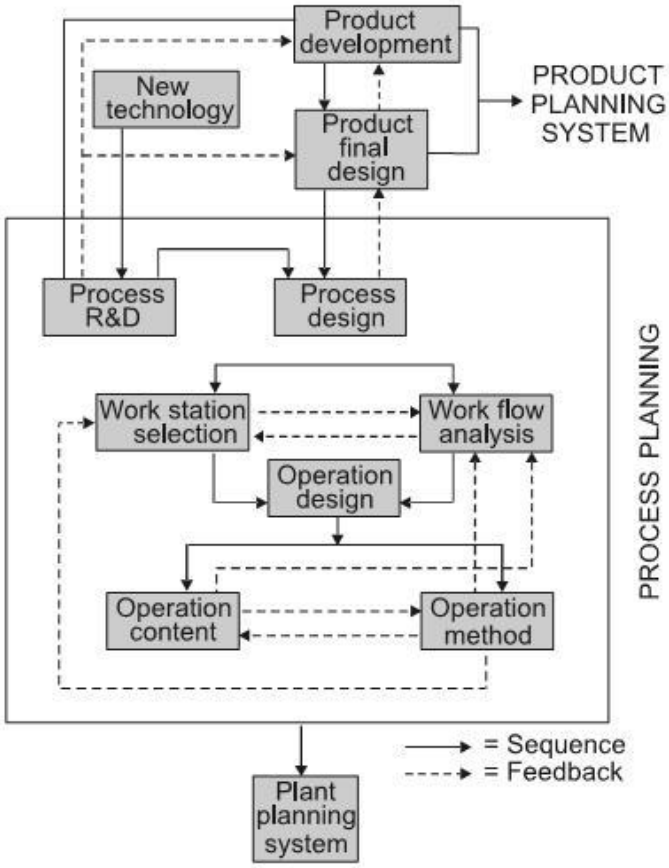
As the process R&D phase progresses, information is feedback to the product development group. If the R&D group will be able to comply with the product specifications, then the product final design stage can begin. If there are problems, however, in complying with product specifications, then modifications in the product development stage must occur. Information from both product final design and process R&D are inputs to the process design

stage. The purpose of this stage is to generate alternative ways of meeting the objectives formalized in the final design stage, determine the criteria by which they will be evaluated, and make the final selection.

Process decisions must be made when

- A new or substantially modified product or service is being offered
- Quality must be improved
- Competitive priorities have changed
- Demand for a product or service is changing
- Current performance is inadequate
- The cost or availability of inputs has changed.

Competitors are gaining by using a new process; or New technologies are available



Process decisions directly affect the process itself, and indirectly the products and services that it provides. Let‘s focus on the relevant common process decisions. In general, Operations managers must consider five common process decisions.

These are:

1. Process choice - whether resources are organized around products or processes. It depends on volume and degree of customization to be provided

2. Vertical integration – backward integration, and forward integration

3. Resource flexibility – ease with which employees and equipment can handle a wide variety of products, output levels, duties, and functions

4. Customer involvement

5. Capital intensity – mix of equipment and human skills in a process

A common classification of production process structures

We often classify processes based on their physical configuration, material and product flow, flexibility, and volume expectation. There are five different process types, which a manager can choose, keeping in mind the relative importance of the following attributes:- Quality, Time, Flexibility, and Cost.

These are:

1. Project process
2. Job process
3. Batch process
4. Line process, and
5. Continuous process

JOB PROCESS

A job process creates the flexibility needed to produce a variety of products or services in significant quantities. Customization is relatively high and volume for any one product or service is low. A job process primarily organizes all like resources around itself; equipment and workers capable of certain types of work are located together. These resources process all jobs requiring that type of work. This process choice creates jumbled flows through the operations as customization is high and most jobs have a different sequence of processing steps. After job process, let‘s analyze:

BATCH PROCESS

A batch process (disconnected flow processes) differs from the job process with respect to volume, variety, and quantity. The primary difference is that volumes are higher because the same or similar products or services are provided repeatedly. Another difference is that a narrower range of products and services is provided.

LINE PROCESS

A line process (repetitive or discrete flow process) lies between the batch and continuous processes on the continuum; volumes are high, and products or services are standardized, which allows resources to be organized around a product or service. There are line flows, with little inventory held between operations. Each operation performs the same process over and over, with little variability in the products or services provided.

CONTINUOUS PROCESS

A continuous process is the extreme end of high-volume, standardized production with rigid line flows. Its name derives from the way materials move through the process. Usually, the primary material, such as liquid, gas, or powder, moves without stopping through the facility. The processes seem more like separate entities than a series of connected operations. The process is often capital intensive and operated round the clock to maximize utilization and to avoid expensive shutdowns and start-ups.

ADVANTAGES AND DISADVANTAGES OF FLOW PROCESSES

The tightly connected configuration, continuous or repetitive transfer of product, narrow product line, and often automated nature of continuous and repetitive flow processes have several advantages over other production structures. The following are the primary advantages and disadvantages of continuous and repetitive flow processes.

Advantages:

1. Equipment can be specified to perform a narrow range of functions very efficiently
2. Jobs can be specialized, so workers can benefit from repetition of a narrow range of tasks performed at any given work station.
3. Material handling can be simplified using efficient but inflexible, fixed location material handling methods, such as conveyers, pipes, and gravity slides
4. Work-in-process (WIP) inventories are small because products move between work stations with little or no waiting and storage time in between
5. Space utilization is efficient because there is no need to store in-process inventories, and material handling is performed using conveyors, pipes, or slides, so the wide aisles required by fork lifts and other mobile machines can be reduced or eliminated.
6. Quality conformance is easier to achieve because with the narrow range of products workers know the quality requirements and how to achieve them, product changeovers and

equipment setups, which are major causes of quality problems, are infrequent, and repetition improves worker skill, so there is less likelihood of errors.

7. Production scheduling and coordination are relatively easy because there are few separate work orders and only work at the first wok station has to be scheduled; work at the other work stations automatically follows in the same sequence.

8. Costs are easy to monitor because all products undergo the same processing and use the same resources in consistent amounts.

Disadvantages

1. The primary disadvantage is that continuous and repetitive flow processes are inflexible. The process can make only products that require the same processing in the same sequence. In addition, once the process has been established, it is expensive to modify its physical configuration to accommodate new products that require different types of processing or a different sequencing of processing stages. Flow processes are also relatively inflexible in terms of volume changes.

2. Initial costs are high because of the specialized equipment used and the substantial work required to design, set up, and balance the workload at each workstation.

3. Work can become tedious and boring for workers unless jobs are well designed and workers are allowed some flexibility through job rotation and cross-training.

4. The production system is extremely vulnerable to unplanned work stoppages due to machine breakdowns, defective components, or worker errors.

Batch flow processes exhibit many of the same advantages, except that equipment and jobs cannot be as specialized, material flow must be more flexible, inter stage inventories are necessary, and greater storage and transport space is needed than for continuous and repetitive processes. However, the disadvantages are less severe because batch processes are more flexible in both product variety and production volumes, work tends to be less tedious, and the system is less vulnerable to shutdowns because some work stations can operate while others are stopped if there are inter stage inventories.

ADVANTAGES AND DISADVANTAGES OF JOB PROCESS ADVANTAGES:

The advantages and disadvantages of job-shop processes are the opposite of those for continuous and repetitive flow processes. The primary advantage of a job shop is its production flexibility. Any product requiring the types of processing that are available in the work centers can be produced. The ability to accommodate different processing times and lot sizes is an especially crucial aspect of flexibility. The other major advantages of job shops are

low initial costs for general-purpose equipment and greater worker satisfaction because of the variety of work performed.

Disadvantages:

The flexibility and lower capital costs for job-shop processes are not free; the following are some corresponding disadvantages.

1. General-purpose equipment is usually less efficient at processing materials.
2. More skilled, higher-paid employees are needed to set up and operate general purpose equipment and to modify work methods to make a variety of products.
3. Less efficient but more flexible material-handling methods, such as fork lifts and hand trucks, are required.
4. Work-in-process inventories are needed to keep the work centers operating during equipment setups, as well as to provide the scheduling flexibility needed to coordinate the variety of products and job processing times.
5. The large in-process inventories and flexible material-handling systems require more space than do flow processes.
6. Quality conformance is difficult because workers must be familiar with a wider range of quality requirements, they perform more product changeovers, and they cannot spend as much time refining their wok methods for any one product.
7. The variability in process sequencing, lot sizes, and processing times, as well as possible uncertainty about order receipts and due dates, make scheduling and coordinating jobs and equipment very complex. These factors, along with the large in-process inventories, result in long throughput times.
8. The variety of products and their processing requirements make it difficult to assign costs to each product, so it is more difficult to determine the profitability of individual products.

CELLULAR PROCESSES

Organizations often capture some of the efficiencies of flow processes and the flexibility of job-shop processes by creating hybrids of the two, called cellular processes. A cellular process can be thought of as a mixture of mini flow processes, called work cells (or cells), and a job-shop operation. The work cells may perform only two or three activities in a spatially connected flow process, or they may perform several activities connected in sequence. Cellular processes are most commonly used as substitutes for job-shop processes that need increased productivity. Increasingly, however, they are being used in place of flow processes to obtain greater flexibility. They are also becoming a popular way to organize

service operations. To create a cellular process, an organization divides its products into families or group of products that require similar processing steps in the same sequence. A work call is then created to perform these steps in the designated sequence for all the products in the family.

The output of the cell may be a finished product or a semi finished product that must be sent elsewhere for further processing. Some products will not be appropriate for any cell, and many products cannot be made entirely at a single cell, so there will normally be a job-shop subsystem (cell) that can do all the processing steps in any sequence.

ADVANTAGES AND DISADVANTAGES OF CELLULAR PROCESSES

Advantages:

1. Material handling and transport are reduced because the work stations (machines) are spatially close and often are operated as repetitive flow processes.
2. Setup times are reduced because jobs processed at the same cell often have similar characteristics that require less changeover from job to job.
3. Throughput time is reduced because the wait between production stages, the wait for transport, and the transport time are reduced.
4. In-process inventories are smaller because of more efficient scheduling and reduced setup time disruptions. Also, the shorter throughput times reduce the amount of safety stock needed.
5. Less space is needed because the machines in cells are located close together and less in process inventory must be stored.
6. Although some investment in equipment is often required, total equipment costs often decrease because the increase in efficiency and machine utilization means that less total equipment is needed to produce the same amount of output.
7. Workers enjoy more satisfaction because they have greater job variety than workers in either flow processes or job shops, since their work often involves several machines and tasks.
8. Quality improves because of greater job satisfaction, simpler machine setups, and similarity of products within cells, which produce fewer mistakes.

Disadvantages:

Successful implementations of a cellular production system requires a considerable amount of work and expertise to characterize and classify products and then design the appropriate work cells and remaining job-shop process.

PROCESS DESIGN : MACRO

Process design : macro is composed of two aspects: the choice of work station and the choice of work low. Work station selection involves the choice of machines to be included in the process, whereas work flow analysis concerns the flow of work between these stations. It is here that the decision is made as to whether the process will be continuous, intermittent, or some combination of both. In continuous processing, the process is in constant operation and usually involves a high capital-to labour ratio. Typical of a continuous-process orientation are the automobile industry, chemical processing, plastics, some high-volume electronic manufacturers, and some utilities such as telephone, power and gas transmission. Also characteristic of continuous- process industries is a product layout where all the work stations are devoted exclusively to a single product and are grouped according to the processing requirements of that product. In more cases than not, the machines found at these work stations are special-purpose and costly and have little versatility outside their own product line. Work flow is largely specified by the physical characteristics that the product layout takes. Often a conveyor system is used. An example of this kind of process is an assembly line for machining engine blocks at an automotive plant. At the other end of the spectrum we find intermittent processing. Here production or service for any one job is carried on intermittently, not continuously. There is a high mix of products which use the facility, and portions of the process may be in operation several times during the day or only occasionally during the month. Usually the relatively low demand for each product or service does not warrant the high investment in a continuous process. Typical of intermittent processes are job shops, emergency rooms in hospitals, hospital laboratories, most office work, many educational processes, and most services. The predominant plant layout that one expects in intermittent processing is a process layout. In a process layout, machines or services of the same category are grouped together. We therefore find lathes, milling machines, inspection stations, and so on, in one location. A characteristic of the process layout is that the particular sequence of operations that one job follows is seldom repeated by other jobs flowing through the process. For example, the sequence from admittance to discharge in a hospital is seldom the same from one patient to the next. A patient may or may not have X-rays, may or may not have surgery, may or may not undergo physical therapy, etc. Consequently the work flow is not nearly as predictable as in the continuous-process case, and we therefore find variable-path equipment, such as forklift trucks, employed in processes of this type.

The criteria used to make the choice between these two layouts include investment costs, material handling costs, direct labour costs, space requirements, equipment flexibility in

meeting changes in output quantity, system reliability and maintenance costs. These economic and non- economic factors must, in turn, be weighed before making the final decision.

6.4.3 PROCESS DESIGN : MICRO

In the next stage attention shifts to the details that make up the work at each station. Concern is with the operational content and operational method of The task Operational content focuses on the appropriate combination of steps that should be assigned to a work station. Operational method, on the other hand, is concerned with the efficient execution of these steps.

PLANT-PLANNING SYSTEM

If a new plant will be built, then process planning proceeds in a relatively constraint-free setting. If, however, the process must be installed in existing plant, then a set of special constraints must be observed. Therefore, the process planning system must interact with the plant-planning system to ensure feasible layouts.

PROCESS DESIGN AS AN ONGOING ACTIVITY

Process design is not strictly limited to new design. Whenever the costs of inputs change by a substantial amount, or whenever output levels or quality objectives change, process review should be initiated. If the reason for the reexamination is a price change in an input factor, then a new input mix reflecting the different price ratio of the substitute inputs may be desirable. For example, if wage rates increase substantially as a result of a new labour contract, it might be possible that automated sequences in the production line which were inefficient uses of capital before might now be profitable.

Process design, then, should be a continuous activity—not one which is precipitated only when new products or services are introduced. Since it is continuous, the question of how much money to spend in its pursuit is of utmost importance. The amount which should be spend depends, of course, upon the benefits accrued. At the limit, effort should continue as long as the marginal benefits from the improvement are greater than the marginal costs. Again we use the word marginal, since it is only those costs that change which are relevant. Fixed costs and some semi variable costs, to the extent that they are not affected by the redesign, are irrelevant.

For firms at the continuous-process end of the spectrum, considerable effort can be channeled in this direction. Small improvements made in the process are magnified when the volume going through each stage is large. Cost savings in the automotive industry, for example, are measured in fractions of a cent per unit. In the service industries, on the other hand, savings are usually measured in dollars, but the volume through each station is, of course, much less.

If redesign requires investment, it seems reasonable to subject the investment to the review of the capital-budgeting process. Therefore, this investment, along with all the other which the firm is considering, must face the final selection process in which only the most profitable survive.

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UNIT – IV MATERIALS MANAGEMENT

Materials Management – Objectives, Planning, Budgeting and Control. Overview of Materials Management Information Systems (MMIS). Purchasing – Objectives, Functions, Policies, Vendor rating and Value Analysis. Stores Management – Nature, Layout, Classification and Coding. Inventory – Objectives, Costs and control techniques. Overview of JIT.

Why Materials Management?

□□Materials is one of the five M‘s(Men, Material, Money, Machine and Methods) of an industrial organization.

□□Materials offer considerable scope for improving profit.

□□Materials form an important form of current assets in any organization.

□□Value addition is the margin between the raw material value and finished goods value.

□□Suppliers and materials management account for more than 50 percent of total value.

□□Quality of the finished product depends on quality of materials used.

□□Conservation of materials and their availability for posterity is one of the planks of social responsibility of business.

□□Exploring new sources of supply is a challenge for material management executives.

Materials Management

□□Material management involves organizing and coordinating all management functions that are responsible for every aspect of materials, storage, and transformation.

□□Buying, storage, and movement of material are the three basic objectives of materials management.

□□Optimum investment in inventory is the prime objective of materials management**.**

□□Development of personnel is very important for long-term growth of a firm.

□□Engineering groups are primarily responsible for standards of specifications.

Importance of Materials Management

1. Lower prices for materials and equipment
2. Faster inventory turnover
3. Continuity of supply
4. Reduced lead time
5. Reduced transportation costs
6. Less duplication of efforts
7. Elimination of buck-passing
8. Reduced materials obsolescence

9. Improved supplier relationships and better records, and information

10. Better interdepartmental cooperation

11. Personnel development

Functions of Materials Management

1. Materials planning and programming

2. Raw material purchase

3. Receiving, store keeping, and warehousing

4. Issuing of material

5. Inventory control

6. Value engineering

7. Transportation of materials

8. Vendor development

9. Vendor rating

10. Disposal of scrap and surpluses

Objectives

Primary Objectives

1. Buying the best item at the lowest cost

2. Reduction in inventory cost and High inventory turnover

3. Maintaining the flow of production

4. Maintaining the consistency of quality

5. Optimization of acquisition and possession, resulting in lower cost

6. Cordial relationship with suppliers

7. Maintaining good records

8. Contribution towards competitiveness

9. Personnel development

Secondary Objectives

1. Promotion of standardization with suppliers

2. Development of reciprocal relations with customers

3. Committees to decide on economic make –or- buy decisions

4. Development of inter departmental relationships

Material Planning And Control

Material planning is a scientific technique of determining in advance the requirements of raw materials, ancillary parts and components, spares etc. as directed by the production programme.

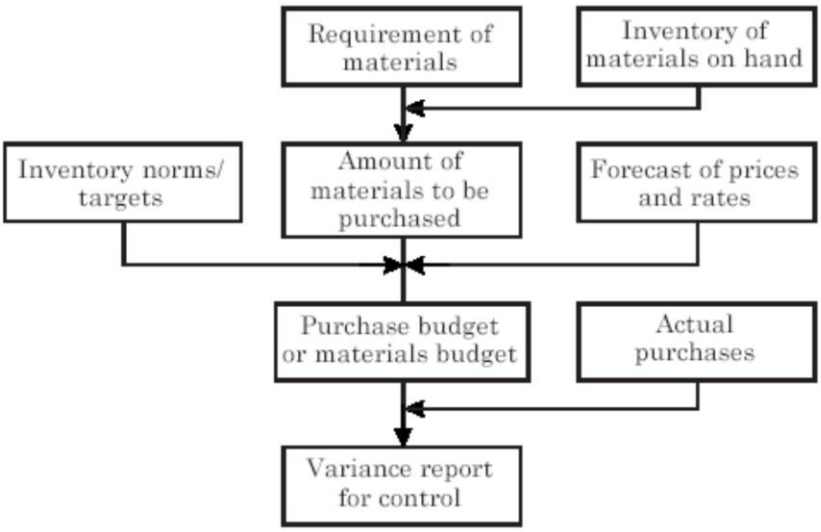
It is a subsystem in the overall planning activity. There are many factors, which influence the activity of material planning. These factors can be classified as macro and micro systems.

1. *Macro factors*: Some of the micro factors which affect material planning, are price trends, business cycles Govt. import policy etc.

2. *Micro factors*: Some of the micro factors that affect material planning are plant capacity utilization, rejection rates, lead times, inventory levels, working capital, delegation of powers and communication.

Materials Budgeting

The Process of Preparing Material Budgets



Materials Budgeting: Process of preparing materials budget or purchase budget in terms of quantity and money value of materials to be procured for a given period of time.

Material Budgeting is an estimate of expenses to be incurred in the procurement of material and its helps in effective execution and control of material plans.

Material Control

Function of maintaining constantly availability of all kinds of materials required for the manufacture of products.

Materials Management Information System (MMIS)

1. MMIS provides on-line information on stock level, work-in- process; finished goods, and stores and spares.
2. It provides information at right time, it also request the supplier to supply the material on time.
3. It also helps in accessing information rapidly, detects errors and assures prompt decision making.
4. It also takes decisions regarding when to buy or when to make parts and the like.

Purchasing

The term _purchasing‘ refer to buying of a material or an item from a company or division that supplies materials. Since a manufacturing firm is involved in the conversion of raw material into finished goods, it should ensure that right type of material are purchased in the right time. A one
percent saving in cost is equivalent to 10 percent increase in turnover.

Objectives of Purchasing

The basic objective of the purchasing function is to ensure continuity of supply of raw materials, sub-contracted items and spare parts and to reduce the ultimate cost of the finished goods. In other words, the objective is not only to procure the raw materials at the lowest price but to reduce the cost of the final product.

The objectives of the purchasing department can be outlined as under:

 To avail the materials, suppliers and equipments at the minimum possible costs: These are the inputs in the manufacturing operations. The minimization of the input cost increases the productivity and resultantly the profitability of the operations.

 To ensure the continuous flow of production through continuous supply of raw materials, components, tools etc. with repair and maintenance service.

 To increase the asset turnover: The investment in the inventories should be kept minimum in relation to the volume of sales. This will increase the turnover of the assets and thus the profitability of the company.

 To develop an alternative source of supply: Exploration of alternative sources of supply of materials increases the bargaining ability of the buyer, minimisation of cost of materials and increases the ability to meet the emergencies.

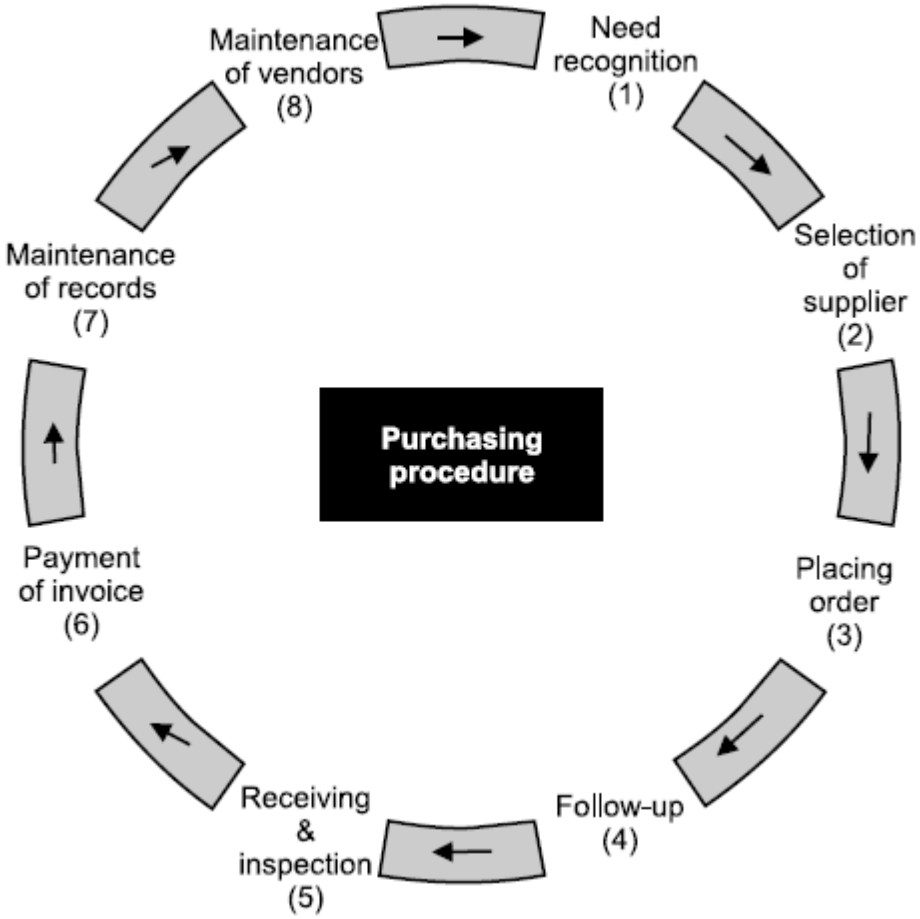
 To establish and maintain the good relations with the suppliers: Maintenance of good relations with the supplier helps in evolving a favourable image in the business circles. Such relations are beneficial to the buyer in terms of changing the reasonable price, preferential allocation of material in case of material shortages, etc.

To achieve maximum integration with other department of the company: The purchase function is related with **production department** for specifications and flow of material, **engineering department** for the purchase of tools, equipments and machines, **marketing department** for the forecasts of sales and its impact on procurement of materials, **financial department** for the purpose of maintaining levels of materials and estimating the working capital required, **personnel department** for the purpose of manning and developing the personnel of purchase department and maintaining good vendor relationship.

 To train and develop the personnel: Purchasing department is manned with varied types of personnel. The company should try to build the imaginative employee force through training and development.

 Efficient record keeping and management reporting: Paper processing is inherent in the purchase function. Such paper processing should be standardised so that record keeping can be facilitated. Periodic reporting to the management about the purchase activities justifies the independent existence of the department.

Purchase Cycle



Purchasing cycle comprising of eight steps.

1. Recognition of Need : Identifying an item that is officially brought to the attention of purchasing department.

2. Description of Requirement: Purchase requisition should give accurate information for ordering recognised product.

3. Selection of Source: Registered suppliers who are approved by the company. Buying from single supplier helps develop long-term relationship and reduces the risk and interruption in the supply. Buying from multiple suppliers may not help in maintaining quality and helps get material at competitive prices.

4. Determination of Price and Availability

5. Placing the Order

6. Order Acknowledgement

7. Follow Up and Expediting

8. Checking The Invoice and Approval

Purchase Functions

1. Obtaining prices
2. Selecting vendors
3. Awarding purchase orders
4. Follow up on delivery promise
5. Adjusting and settling complaints
6. Selecting and training of purchase personnel
7. Vendor relations

Purchasing Policies

• **Ancillary Development:** The firms sub-contracts, i.e., decides to buy the parts from outside suppliers. Mostly the fabricated parts, components are brought from outside suppliers by the firms.

• **Make or buy:** Another purchasing policy is whether to buy the parts or components from outside supplier or manufacture within the firm. The decisions lie depending in various factors.

• **Speculative buying:** Speculative buying is conducted with the hope of making profit out of price changes. Here the profit is made by buying at low price and selling at higher price.

Vendor rating:

The evaluation of supplier or vendor rating provides valuable information which help in improving the quality of the decision. In the vendor rating three basic aspects are considered namely quality, service and price

The Development Project Committee of the National Association of Purchasing Agents (U.S.A.) has suggested following methods for evaluating the performance of past suppliers.

1. The categorical plan: Under this method the members of the buying staff related with the supplier like receiving section, quality control department, manufacturing department etc., are required to assess the performance of each supplier. The rating sheets are provided with the record of the supplier, their product and the list of factors for the evaluation purposes. The members of the buying staff are required to assign the plus or minus notations against each factor.

2. The weighted-point method: The weighted-point method provides the quantitative data for each factor of evaluation. The weights are assigned to each factor of evaluation according to the need of the organization, *e.g.*, a company decides the three factors to be considered—quality, price and timely delivery. It assigns the relative weight to each of these factors as under:

Quality …… 50 points

Price …… 30 points

Timely delivery …… 20 points

3. Critical incidents method: Record of events related to buyer vendor relationships is maintained in each vendor‘s file. They reflect positive and negative aspect of actual performance. This kind of documentation useful in discussing ways and means of improving performance, acknowledging the existence of good relationships, determining the competence of a vendor, and if necessary considering termination

4. The cost-ratio plan: Under this method, the vendor rating is done on the basis of various costs incurred for procuring the materials from various suppliers. The cost-ratios are ascertained delivery etc. The cost-ratios are ascertained for the different rating variables such as quality, price, timely delivery etc. The cost-ratio is calculated in percentage on the basis of total individual cost and total value of purchases.

5. Checklist system: A simple checklist is used to evaluate the vendors. Check list may be something like Reliability, technical capability, after sales service, availability, buying convenience etc

Value Analysis

Value engineering or value analysis had its birth during the World War II Lawrence D. Miles was responsible for developing the technique and naming it. Value analysis is defined as ―an organized creative approach which has its objective, the efficient identification of unnecessary cost-cost which provides neither quality nor use nor life nor appearance nor customer features.‖

Value analysis focuses engineering, manufacturing and purchasing attention to one objective equivalent performance at a lower cost.

Value = Function / Cost

Steps in Value Analysis

In order to answer the above questions, three **basic steps** are necessary:

1. **Identifying the function:** Any useful product has some primary function which must be identified—a bulb to give light, a refrigerator to preserve food, etc. In addition it may have secondary functions such as withstanding shock, etc. These two must be identified.

2. **Evaluation of the function by comparison:** Value being a relative term, the comparison approach must be used to evaluate functions. The basic question is, _Does the function accomplish reliability at the best cost‘ and can be answered only comparison.

3. **Develop alternatives:** Realistic situations must be faced, objections should overcome and effective engineering manufacturing and other alternatives must be developed.

Stores Management

Stores play a vital role in the operations of company. It is in direct touch with the user departments in its day-to-day activities. The most important purpose served by the stores is to provide uninterrupted service to the manufacturing divisions. Further, stores are often equated directly with money, as money is locked up in the stores.

Nature of Stores

□□Store as building where inventories are kept.

□□Storage is the function of receiving, storing, and issuing materials.

□□Stores ensure ready accessibility of major materials there-by efficient service to users.

□□Minimisation of stores cost, and continuous supply is the prime function of stores.

□□Stores layout is a fundamental factor in determining the efficient performance of stores department.

□□A satisfactory storage system compromises between the use of space and the use of time.

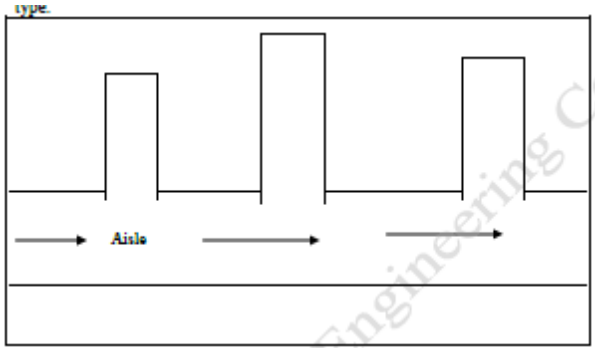
□□Random location means that items can be stored in any storage position which is available.

□□Keeping stock on one side of the aisle in which case the layout is called comb type .

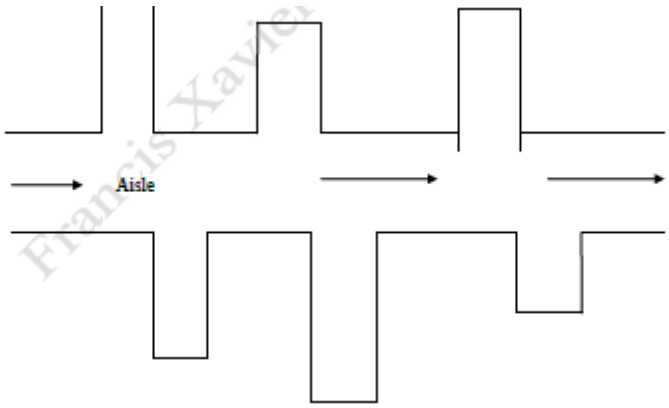
□□Stores manual is a written statement of policies, and procedures.

Layout:

Comb type layout: Stock may be kept on one side of the aisle in which case it is called comb type.



Tree type layout: Goods may be placed on either side of the aisle in which case, the method is called tree type layout



CLASSIFICATION AND CODIFICATION:

It is one of the functions of stores management. Codification is a process of representing each item by a number, the digit of which indicates the group, the sub-group, the type and the dimension of the item.

OBJECTIVES OF CODIFICATION

The objectives of a rationalized material coding system are:

1. Bringing all items together.
2. To enable putting up of any future item in its proper place.
3. To classify an item according to its characteristics.
4. To give an unique code number to each item to avoid duplication and ambiguity.
5. To reveal excessive variety and promote standardization and variety reduction.
6. To establish a common language for the identification of an item.
7. To fix essential parameters for specifying an item.
8. To specify item as per national and international standards.
9. To enable data processing and analysis.

Inventory

A physical resource that a firm holds in stock with the intent of selling it or transforming it into a more valuable state.

□□Raw Materials

□□Works-in-Process

□□Finished Goods

□□Maintenance, Repair and Operating (MRO)

Objectives:

The main objective of inventory management is to maintain inventory at appropriate level to avoid excessive or shortage of inventory because both the cases are undesirable for business.

Thus, management is faced with the following conflicting objectives:

1. To keep inventory at sufficiently high level to perform production and sales activities smoothly.
2. To minimize investment in inventory at minimum level to maximize profitability.

Other objectives of inventory management are explained as under:-

1. To ensure that the supply of raw material & finished goods will remain continuous so that production process is not halted and demands of customers are duly met.
2. To minimize carrying cost of inventory.
3. To keep investment in inventory at optimum level.

4. To reduce the losses of theft, obsolescence & wastage etc.

5. To make arrangement for sale of slow moving items.

6. To minimize inventory ordering costs.

Inventory Control:

Inventory Control is the supervision of supply, storage and accessibility of items in order to ensure an adequate supply without excessive oversupply.

It can also be referred as internal control - an accounting procedure or system designed to promote efficiency or assure the implementation of a policy or safeguard assets or avoid fraud and error etc.

Objectives of Inventory Control

1. To ensure adequate supply of products to customer and avoid shortages as far as possible.

2. To make sure that the financial investment in inventories is minimum (*i.e.,* to see that the working capital is blocked to the minimum possible extent).

3. Efficient purchasing, storing, consumption and accounting for materials is an important objective.

4. To maintain timely record of inventories of all the items and to maintain the stock within the desired limits

5. To ensure timely action for replenishment.

6. To provide a reserve stock for variations in lead times of delivery of materials.

7. To provide a scientific base for both short-term and long-term planning of materials.

Inventory Costs

Inventory costs includes ordering cost plus carrying costs.

Ordering Cost

Cost of procurement and inbound logistics costs form a part of Ordering Cost. Ordering Cost is dependant and varies based on two factors - The cost of ordering excess and the Cost of ordering too less.

Both these factors move in opposite directions to each other. Ordering excess quantity will result in carrying cost of inventory. Where as ordering less will result in increase of replenishment cost and ordering costs.

These two above costs together are called Total Stocking Cost. If you plot the order quantity vs the TSC, you will see the graph declining gradually until a certain point after which with every increase in quantity the TSC will proportionately show an increase.

This functional analysis and cost implications form the basis of determining the Inventory Procurement decision by answering the two basic fundamental questions - How Much to Order and When to Order.

How much to order is determined by arriving at the Economic Order Quantity or EOQ.

Carrying Cost

Inventory storage and maintenance involves various types of costs namely:

□□Inventory Storage Cost

□□Cost of Capital

Inventory carrying involves Inventory storage and management either using in house facilities or external warehouses owned and managed by third party vendors. In both cases, inventory management and process involves extensive use of Building, Material Handling Equipment, IT Software applications and Hardware Equipment coupled managed by Operations and Management Staff resources.

Inventory Storage Cost

Inventory storage costs typically include Cost of Building Rental and facility maintenance and related costs. Cost of Material Handling Equipment, IT Hardware and applications, including cost of purchase, depreciation or rental or lease as the case may be. Further costs include operational costs, consumables, communication costs and utilities, besides the cost of human resources employed in operations as well as management.

Cost of Capital

Includes the costs of investments, interest on working capital, taxes on inventory paid, insurance costs and other costs associate with legal liabilities.

The inventory storage costs as well as cost of capital is dependent upon and varies with the decision of the management to manage inventory in house or through outsourced vendors and third party service providers.

Inventory Control Techniques:

Economic Order Quantity:

EOQ (Economic Order Quantity) -EOQ is the level of inventory order that minimizes the total cost associated with inventory management. The objective is to find out and maintain optimum level of investment in inventory to minimize the total cost associated with it.

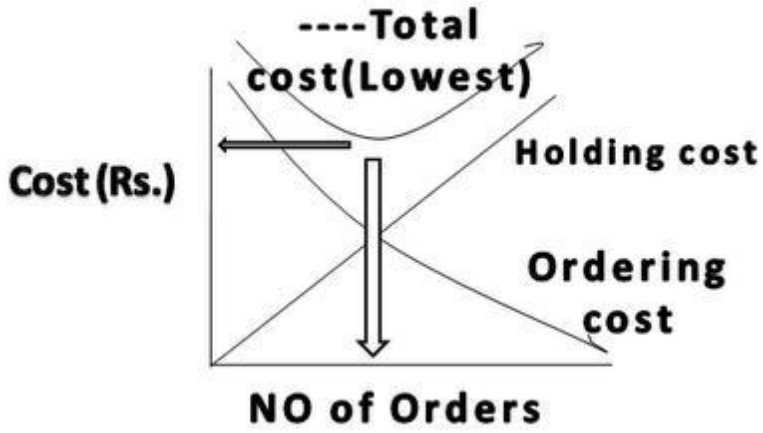
The total cost includes:

Carrying Cost: Are associated with the maintenance/holding of inventory.

Ordering Cost: Are costs associated with acquisition of/placing order for inventory.

$$EOQ = \sqrt{\frac{2 \times \text{Annual Consumption} \times \text{Ordering Cost}}{\text{Storage(holding)cost per unit}}}$$

$$\text{Storage (holding) cost per unit} = \text{Cost per unit} \times \text{Storage cost(\%)}$$



It is that quantity where the total of ordering cost and holding cost remains lowest.

Re-order level: stock level at which fresh order is placed.

Average consumption per day x lead time + buffer stock

Lead time: Duration time between placing an order & receipt of material

Ideal – 2 to 6 weeks.

1. **Always better control (ABC) classification.** In this analysis, the classification of existing inventory is based on annual consumption and the annual value of the items. Hence we obtain the quantity of inventory item consumed during the year and multiply it by unit cost to obtain annual usage cost. The items are then arranged in the descending order of such

annual usage cost.

(*a*) ***A-Item:*** Very tight control, the items being of high value. The control need be exercised at higher level of authority.

A ITEMS

Small in number, but consume large amount of resources

Must have:

• Tight control

• Rigid estimate of requirements

• Strict & closer watch

• Low safety stocks

• Managed by top management

B-Item: Moderate control, the items being of moderate value. The control need be exercised at middle level of authority.

B ITEM

Intermediate

Must have:

• Moderate control

• Purchase based on rigid requirements

• Reasonably strict watch & control

• Moderate safety stocks

• Managed by middle level management

(*c*) ***C-Item:*** The items being of low value, the control can be exercised at gross root level of authority, *i.e*., by respective user department managers.

C ITEMS

Larger in number, but consume lesser amount of resources

Must have:

• Ordinary control measures

• Purchase based on usage estimates

• High safety stocks

ABC analysis does not stress on items those are less costly but may be vital

This is based on cost criteria.

It helps to exercise selective control when confronted with large number of items it rationalizes the number of orders, number of items & reduce the inventory.

About 10 % of materials consume 70 % of resources

About 20 % of materials consume 20 % of resources

About 70 % of materials consume 10 % of resources

2.High, medium and low (HML) classification. In this analysis, the classification of existing inventory is based on unit price of the items. They are classified as high price, medium price and low cost items.

3.Vital, essential and desirable (VED) classification. In this analysis, the classification of existing inventory is based on criticality of the items. They are classified as vital, essential and desirable items. It is mainly used in spare parts inventory.

4.Scarce, difficult and easy to obtain (SDE). In this analysis, the classification of existing inventory is based on the items.

5.GOLF analysis: In this analysis, the classification of existing inventory is based sources of the items. They are classified as Government supply, ordinarily available, local availability and foreign source of supply items.

6.SOS analysis: In this analysis, the classification of existing inventory is based nature of supply of items. They are classified as seasonal and off-seasonal items.

7.Fast moving, slow moving and non-moving (FSN).

8.Economic order quantity (EOQ). Inventory models deal with idle resources like men, machines, money and materials. These models are concerned with two decisions: how much to order (purchase or produce) and when to order so as to minimize the total cost.

9.Max-Minimum system.

10.Two bin system

Overview of JIT

What is a "Just-in-time System"?

Just-in-time‖: A philosophy of manufacturing based on planned elimination of all waste and continuous improvement of productivity.

Just in time is a _pull' system of production, so actual orders provide a signal for when a product should be manufactured. Demand-pull enables a firm to produce only what is required, in the correct quantity and at the correct time.

This means that stock levels of raw materials, components, work in progress and finished goods can be kept to a minimum. This requires a carefully planned scheduling and flow of resources through the production process. Modern manufacturing firms use sophisticated production scheduling software to plan production for each period of time, which includes ordering the correct stock. Information is exchanged with suppliers and customers through EDI (Electronic

Data Interchange) to help ensure that every detail is correct. Supplies are delivered right to the production line only when they are needed. For example, a car manufacturing plant might receive exactly the right number and type of tyres for one day's production, and the supplier would be expected to deliver them to the correct loading bay on the production line within a very narrow time slot.

Overview of JIT manufacturing

1. Inventory reduction
2. Quality improvement
3. Lead time reduction
4. Lead time reduction

5. Continuous Improvement
6. Total Preventive Maintenance
7. Strategic Gain

Advantages of JIT

□□Lower stock holding means a reduction in storage space which saves rent and insurance costs

□□As stock is only obtained when it is needed, less working capital is tied up in stock

□□There is less likelihood of stock perishing, becoming obsolete or out of date

□□Avoids the build-up of unsold finished product that can occur with sudden changes in demand

□□Less time is spent on checking and re-working the product of others as the emphasis is on getting the work **right first time**

Disadvantages of JIT

□□There is little room for mistakes as minimal stock is kept for re-working faulty product

□□Production is very reliant on suppliers and if stock is not delivered on time, the whole production schedule can be delayed

□□There is no spare finished product available to meet unexpected orders, because all product is made to meet actual orders – however, JIT is a very responsive method of production.

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UNIT – V PROJECT AND FACILITY PLANNING

Project Management – Scheduling Techniques, PERT, CPM, Crashing CPM networks – Simple Problems. Facility Location – Theories, Steps in Selection, Location Models – Simple Problems. Facility Layout – Principles, Types, Planning tools and techniques.

Project Management:

A project is an organized endeavour to accomplish a specified non-routine or low volume task. Although projects are not repetitive, they take significant amount of time to complete and are large scale or complex enough to be recognized and managed as separate undertakings.

Project life cycle:

1. The concept phase: During this phase, the organization realises that a project may be needed or the organization is requested to propose a plan to perform a project for some customer.
2. Initial planning or feasibility phase: During the phase, the project manager plans the project to a level of detail, sufficient for initial scheduling and budgeting.
3. Detailed planning phase: If the project is approved, then detailed scheduling and budgeting is done in this phase.
4. Organisation phase: During this phase, a detailed project definition such as the work breakdown structure is examined.
5. Execution phase: During this phase the various activities planned and completed as per the schedule, utilizing the allotted resources.
6. Termination phase: This is the phase, during which the project is terminated or disbanded after completion. The personnel who were working in the project are assigned back to their regular jobs or to other jobs in the organisation or to other projects in this phase

Role of a Project Manager:

1. Manage the project‘s scope to define the goals and work to be done, in sufficient detail to facilitate understanding and correct performance by the participants
2. Manage the human resources involved in the project
3. Manage communications to see that, the appropriate parties are informed
4. Manage time by planning and meeting a schedule
5. Manage quality so that, the projects results are satisfactory
6. Manage costs so that, the project is performed at the minimum practical cost and with in budget if possible.

Terms Used in Project Management

Activity : A certain amount of work or task required in the project

Activity duration: In CPM the best estimate of time to complete an activity . In PERT the expected time or average time to complete an activity

Critical activity : An activity that has no room for schedule slippages **:** if it slips the entire the entire project completion will slip. An activity with zero slack

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Event :A beginning , a completion point ,or a milestone accomplishment within the project. An activity begins and ends with events

Latest finish (LF) : The latest that an activity can finish from the beginning of the project

Latest start (LS) :The latest that an activity can start from the beginning of the project

Most likely time (t m) : The time for completing the activity that is is the consensus best estimate, used in PERT

Optimistic Time (to): The time for completing an activity if all goes well : used in PERT

Pessimistic Time (tp): The time for completing an activity if bad luck is encountered : used in PERT

Predecessor activity : An activity that must occur before another activity .

Slack : The amount of time that an activity or group of activities can slip without causing a delay in the completion of the project

Successor activity : An activity that must occur after another activity

Conventions used in drawing network diagrams (Arrows & Circles)

 Activity on Arrow (AOA) : The activities are denoted by Arrows and events are denoted by circles

 Activity on Node(AON) : Activities are denoted by circles(or nodes) and the precedence relation ships between activities are indicated by arrows

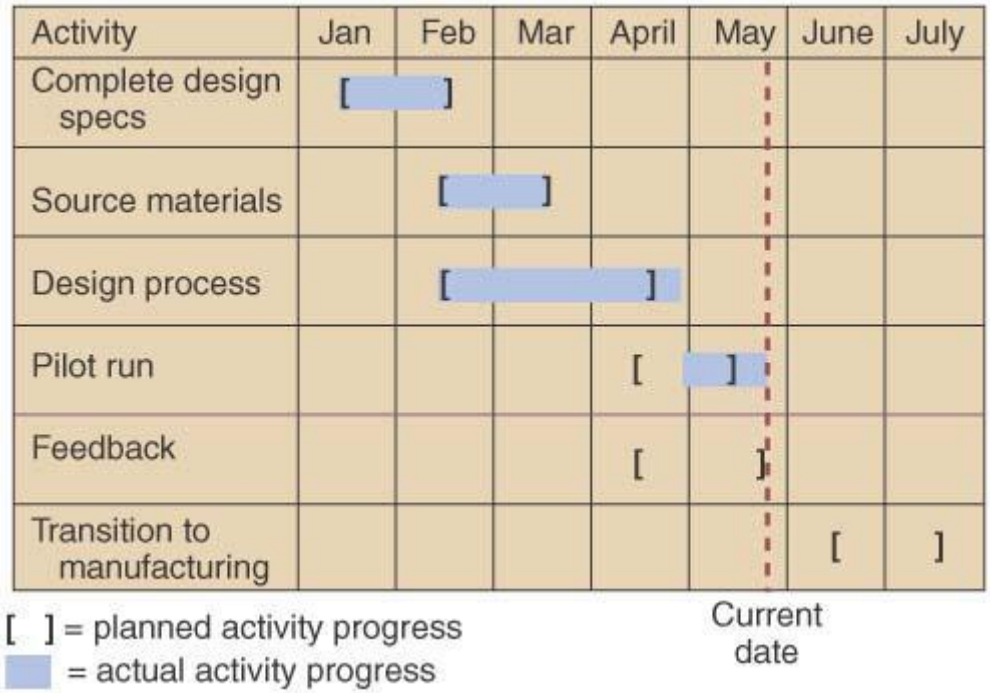
Scheduling:

Scheduling is the process of deciding how to commit resources between a variety of possible tasks. Time can be specified (scheduling a flight to leave at 8:00) or floating as part of a sequence of events.

Scheduling Techniques

Gantt Chart:

During the era of scientific management, Henry Gantt developed a tool for displaying the progression of a project in the form of a specialized chart. An early application was the tracking of the progress of ship building projects. Today, Gantt's scheduling tool takes the form of a horizontal bar graph and is known as a Gantt chart, a basic sample of which is shown below:



The horizontal axis of the Gantt chart is a time scale, expressed either in absolute time or in relative time referenced to the beginning of the project. The time resolution depends on the project - the time unit typically is in weeks or months. Rows of bars in the chart show the beginning and ending dates of the individual tasks in the project.

In the above example, each task is shown to begin when the task above it completes. However, the bars may overlap in cases where a task can begin before the completion of another, and there may be several tasks performed in parallel. For such cases, the Gantt chart is quite useful for communicating the timing of the various tasks.

For larger projects, the tasks can be broken into subtasks having their own Gantt charts to maintain readability.

Gantt Chart Enhancements

This basic version of the Gantt chart often is enhanced to communicate more information.

A vertical marker can used to mark the present point in time.

The progression of each activity may be shown by shading the bar as progress is made, allowing the status of each activity to be known with just a glance.

Dependencies can be depicted using link lines or color codes.

Resource allocation can be specified for each task.

Milestones can be shown.

PERT:

Complex projects require a series of activities, some of which must be performed sequentially and others that can be performed in parallel with other activities. This collection of series and parallel tasks can be modeled as a network. In 1957 the Critical Path Method (CPM) was developed as a network model for project management. CPM is a deterministic method that uses a fixed time estimate for each activity.

While CPM is easy to understand and use, it does not consider the time variations that can have a great impact on the completion time of a complex project.

The Program Evaluation and Review Technique (PERT) is a network model that allows for randomness in activity completion times. PERT was developed in the late 1950's for the U.S.

Navy's Polaris project having thousands of contractors. It has the potential to reduce both the time and cost required to complete a project.

The Network Diagram

In a project, an activity is a task that must be performed and an event is a milestone marking the completion of one or more activities. Before an activity can begin, all of its predecessor activities must be completed. Project network models represent activities and milestones by arcs and nodes. PERT originally was an activity on arc network, in which the activities are represented on the lines and milestones on the nodes. Over time, some people began to use PERT as an activity on node network. For this discussion, we will use the original form of activity on arc.

The PERT chart may have multiple pages with many sub-tasks. The following is a very simple example of a **PERT diagram:**

PERT Chart

The milestones generally are numbered so that the ending node of an activity has a higher number than the beginning node. Incrementing the numbers by 10 allows for new ones to be inserted without modifying the numbering of the entire diagram. The activities in the above diagram are labeled with letters along with the expected time required to complete the activity.

Steps in the PERT Planning Process

PERT planning involves the following steps:

Identify the specific activities and milestones.

Determine the proper sequence of the activities.

Construct a network diagram.

Estimate the time required for each activity.

Determine the critical path.

Update the PERT chart as the project progresses.

1. Identify Activities and Milestones

The activities are the tasks required to complete the project. The milestones are the events marking the beginning and end of one or more activities. It is helpful to list the tasks in a table that in later steps can be expanded to include information on sequence and duration.

2. Determine Activity Sequence

This step may be combined with the activity identification step since the activity sequence is evident for some tasks. Other tasks may require more analysis to determine the exact order in which they must be performed.

3. Construct the Network Diagram

Using the activity sequence information, a network diagram can be drawn showing the sequence of the serial and parallel activities. For the original activity-on-arc model, the activities are depicted by arrowed lines and milestones are depicted by circles or "bubbles".

If done manually, several drafts may be required to correctly portray the relationships among activities. Software packages simplify this step by automatically converting tabular activity information into a network diagram.

4. Estimate Activity Times

Weeks are a commonly used unit of time for activity completion, but any consistent unit of time can be used.

A distinguishing feature of PERT is its ability to deal with uncertainty in activity completion times. For each activity, the model usually includes three time estimates:

Optimistic time - generally the shortest time in which the activity can be completed. It is common practice to specify optimistic times to be three standard deviations from the mean so that there is approximately a 1% chance that the activity will be completed within the optimistic time.

Most likely time - the completion time having the highest probability. Note that this time is different from the expected time.

Pessimistic time - the longest time that an activity might require. Three standard deviations from the mean is commonly used for the pessimistic time.

PERT assumes a beta probability distribution for the time estimates. For a beta distribution, the expected time for each activity can be approximated using the following weighted average:

Expected time = (Optimistic + 4 x Most likely + Pessimistic) / 6

This expected time may be displayed on the network diagram.

To calculate the variance for each activity completion time, if three standard deviation times were selected for the optimistic and pessimistic times, then there are six standard deviations between them, so the variance is given by:

$$[(\text{Pessimistic} - \text{Optimistic}) / 6]^2$$

5. Determine the Critical Path

The critical path is determined by adding the times for the activities in each sequence and determining the longest path in the project. The critical path determines the total calendar time required for the project. If activities outside the critical path speed up or slow down (within limits), the total project time does not change. The amount of time that a non-critical path activity can be delayed without delaying the project is referred to as slack time.

If the critical path is not immediately obvious, it may be helpful to determine the following four quantities for each activity:

ES - Earliest Start time

EF - Earliest Finish time

LS - Latest Start time

LF - Latest Finish time

These times are calculated using the expected time for the relevant activities. The earliest start and finish times of each activity are determined by working forward through the network and determining the earliest time at which an activity can start and finish considering its predecessor activities. The latest start and finish times are the latest times that an activity can start and finish without delaying the project. LS and LF are found by working backward through the network.

The difference in the latest and earliest finish of each activity is that activity's slack. The critical path then is the path through the network in which none of the activities have slack.

The variance in the project completion time can be calculated by summing the variances in the completion times of the activities in the critical path. Given this variance, one can calculate the probability that the project will be completed by a certain date assuming a normal probability distribution for the critical path. The normal distribution assumption holds if the number of activities in the path is large enough for the central limit theorem to be applied.

Since the critical path determines the completion date of the project, the project can be accelerated by adding the resources required to decrease the time for the activities in the critical path. Such a shortening of the project sometimes is referred to as project crashing.

6. Update as Project Progresses

Make adjustments in the PERT chart as the project progresses. As the project unfolds, the estimated times can be replaced with actual times. In cases where there are delays, additional resources may be needed to stay on schedule and the PERT chart may be modified to reflect the new situation.

Benefits of PERT

PERT is useful because it provides the following information:

Expected project completion time.

Probability of completion before a specified date.

The critical path activities that directly impact the completion time.

The activities that have slack time and that can lend resources to critical path activities.

Activity start and end dates.

Limitations

The following are some of PERT's weaknesses:

The activity time estimates are somewhat subjective and depend on judgement. In cases where there is little experience in performing an activity, the numbers may be only a guess. In other cases, if the person or group performing the activity estimates the time there may be bias in the estimate.

Even if the activity times are well-estimated, PERT assumes a beta distribution for these time estimates, but the actual distribution may be different.

Even if the beta distribution assumption holds, PERT assumes that the probability distribution of the project completion time is the same as the that of the critical path. Because other paths can become the critical path if their associated activities are delayed, PERT consistently underestimates the expected project completion time.

The underestimation of the project completion time due to alternate paths becoming critical is perhaps the most serious of these issues. To overcome this limitation, Monte Carlo simulations can be performed on the network to eliminate this optimistic bias in the expected project completion time.

CPM:

In 1957, DuPont developed a project management method designed to address the challenge of shutting down chemical plants for maintenance and then restarting the plants once the maintenance had been completed. Given the complexity of the process, they developed the

Critical Path Method (CPM) for managing such projects.

CPM provides the following benefits:

- Provides a graphical view of the project.
- Predicts the time required to complete the project.
- Shows which activities are critical to maintaining the schedule and which are not.

CPM models the activities and events of a project as a network. Activities are depicted as nodes on the network and events that signify the beginning or ending of activities are depicted as arcs or lines between the nodes. The following is an example of a CPM network diagram:

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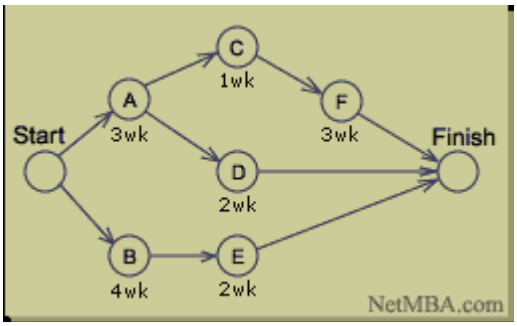
CPM Diagram



Steps in CPM Project Planning

1. Specify the individual activities.
2. Determine the sequence of those activities.
3. Draw a network diagram.
4. Estimate the completion time for each activity.
5. Identify the critical path (longest path through the network)
6. Update the CPM diagram as the project progresses.

1. Specify the Individual Activities

From the work breakdown structure, a listing can be made of all the activities in the project. This listing can be used as the basis for adding sequence and duration information in later steps.

2. Determine the Sequence of the Activities

Some activities are dependent on the completion of others. A listing of the immediate predecessors of each activity is useful for constructing the CPM network diagram.

3. Draw the Network Diagram

Once the activities and their sequencing have been defined, the CPM diagram can be drawn. CPM originally was developed as an *activity on node* (AON) network, but some project planners prefer to specify the activities on the arcs.

4. Estimate Activity Completion Time

The time required to complete each activity can be estimated using past experience or the estimates of knowledgeable persons. CPM is a deterministic model that does not take into account variation in the completion time, so only one number is used for an activity's time estimate.

5. Identify the Critical Path

The critical path is the longest-duration path through the network. The significance of the critical path is that the activities that lie on it cannot be delayed without delaying the project. Because of its impact on the entire project, critical path analysis is an important aspect of project planning.

The critical path can be identified by determining the following four parameters for each activity:

- ES - earliest start time: the earliest time at which the activity can start given that its precedent activities must be completed first.
- EF - earliest finish time, equal to the earliest start time for the activity plus the time required to complete the activity.
- LF - latest finish time: the latest time at which the activity can be completed without delaying the project.
- LS - latest start time, equal to the latest finish time minus the time required to complete the activity.

The *slack time* for an activity is the time between its earliest and latest start time, or between its earliest and latest finish time. Slack is the amount of time that an activity can be delayed past its earliest start or earliest finish without delaying the project.

The critical path is the path through the project network in which none of the activities have slack, that is, the path for which ES=LS and EF=LF for all activities in the path. A delay in the critical path delays the project. Similarly, to accelerate the project it is necessary to reduce the total time required for the activities in the critical path.

6. Update CPM Diagram

As the project progresses, the actual task completion times will be known and the network diagram can be updated to include this information. A new critical path may emerge, and structural changes may be made in the network if project requirements change.